

#1 · Number Sense and Operations

What is a mixed number, and how do you write $2\frac{3}{4}$ as an improper fraction?

Answer

A mixed number has a whole part and a fraction part, like $2\frac{3}{4}$. Multiply the whole number by the denominator ($2 \times 4 = 8$), add the numerator ($8 + 3 = 11$), and keep the same denominator, so the improper fraction is $\frac{11}{4}$.

#2 · Patterns and Relations

The pattern is 5, 9, 13, 17. What is the next term, and what is the rule?

Answer

The next term is 21. The rule is add 4 each time ($17 + 4 = 21$). Each term grows by the same amount, so it is an arithmetic pattern.

#3 · Geometry and Measurement

**A rectangle is 8 cm long and 5 cm wide.
What is its perimeter?**

Answer

The perimeter is 26 cm. Add the length and width and double it: $2 \times (8 + 5) = 2 \times 13 = 26$ cm. The perimeter is the total distance around the shape.

#4 · Data Management and Probability

The numbers are 10, 20, 30, and 40. What is the mean?

Answer

The mean, or average, is 25. Add all the numbers ($10 + 20 + 30 + 40 = 100$), then divide by how many numbers there are (100 divided by $4 = 25$).

#5 · Number Sense and Operations

Simplify the fraction 12/18 to its lowest terms.

Answer

The simplified fraction is $\frac{2}{3}$. Divide the numerator (top number) and denominator (bottom number) by the greatest common factor, which is 6: 12 divided by 6 = 2, and 18 divided by 6 = 3.

#6 · Patterns and Relations

The pattern is 2, 5, 8, 11. What is the 10th term?

Answer

The 10th term is 29. The rule is $3n$ minus 1, where n is the term position. For the 10th term, $3 \times 10 = 30$, then 30 minus $1 = 29$.

#7 · Geometry and Measurement

A triangle has a base of 10 cm and a height of 6 cm. What is its area?

Answer

The area is 30 square cm. Use the formula area = one half x base x height: $0.5 \times 10 \times 6 = 30$. The base and height must meet at a right angle.

#8 · Data Management and Probability

The numbers are 3, 7, 2, 9, and 5. What is the median?

Answer

The median is 5. Put the numbers in order first: 2, 3, 5, 7, 9. With an odd count, the median is the middle number, which is 5.

#9 · Number Sense and Operations

**Add one quarter ($\frac{1}{4}$) and one sixth ($\frac{1}{6}$).
Show the steps.**

Answer

The sum is $\frac{5}{12}$. Find a common denominator of 12, then rewrite each fraction: $\frac{1}{4} = \frac{3}{12}$ and $\frac{1}{6} = \frac{2}{12}$. Add the numerators: $3 + 2 = 5$, so the answer is $\frac{5}{12}$.

#10 · Patterns and Relations

Simplify the ratio 12 to 18. What is the simplest ratio?

Answer

The simplest ratio is 2 to 3, written as 2:3. Divide both numbers by the greatest common factor, which is 6: 12 divided by 6 = 2, and 18 divided by 6 = 3.

#11 · Geometry and Measurement

**A rectangle is 12 m long and 7 m wide.
What is its area?**

Answer

The area is 84 square metres. Multiply the length by the width: $12 \times 7 = 84$. Area measures the space inside a flat shape.

#12 · Data Management and Probability

The numbers are 2, 3, 3, 4, and 5. What is the mode?

Answer

The mode is 3. The mode is the value that appears most often in a list, and 3 appears twice while the others appear once.

#13 · Number Sense and Operations

Write three fifths ($\frac{3}{5}$) as a decimal.

Answer

The decimal is 0.6. Divide the numerator by the denominator: 3 divided by 5 = 0.6. A fraction bar means divide the top number by the bottom number.

#14 · Patterns and Relations

A car travels 240 km in 3 hours. What is its average speed?

Answer

The average speed is 80 km per hour. Divide the distance by the time: 240 divided by 3 = 80. Speed is the distance travelled in one unit of time.

#15 · Geometry and Measurement

A circle has a radius of 7 cm. Using pi as 22/7, what is its circumference?

Answer

The circumference is 44 cm. Use the formula $\text{circumference} = 2 \times \pi \times \text{radius}$: $2 \times (22/7) \times 7 = 2 \times 22 = 44$. Circumference is the distance around the circle.

#16 · Data Management and Probability

The numbers are 12, 25, 8, and 30. What is the range?

Answer

The range is 22. Subtract the smallest value from the largest: $30 \text{ minus } 8 = 22$. The range shows how spread out the data is.

#17 · Number Sense and Operations

Write 45% as a decimal.

Answer

The decimal is 0.45. To convert a percent to a decimal, divide by 100, which means moving the decimal point two places left: 45% becomes 0.45.

#18 · Patterns and Relations

Solve for x: x divided by 6 = 10 divided by 15.

Answer

x equals 4. Cross multiply the two fractions: 15 times x = 6 times 10, so $15x = 60$. Divide both sides by 15 to get $x = 4$.

#19 · Geometry and Measurement

A circle has a radius of 7 cm. Using pi as $\frac{22}{7}$, what is its area?

Answer

The area is 154 square cm. Use the formula $\text{area} = \pi \times \text{radius squared}$: $(\frac{22}{7}) \times 7 \times 7$. The 7s cancel with 7, leaving $22 \times 7 = 154$.

#20 · Data Management and Probability

The values are 10, 12, 14, and 16. How does adding a very large outlier, like 100, affect the mean versus the median?

Answer

The mean rises a lot while the median barely moves. The five values 10, 12, 14, 16, 100 give a mean of 152 divided by 5 = 30.4, but the median stays 14. Outliers pull the mean, so the median is more reliable when a value is extreme.

#21 · Number Sense and Operations

What is 25% of 80?

Answer

The answer is 20. Convert 25% to a decimal (0.25), then multiply by 80: $0.25 \times 80 = 20$. You can also think of 25% as one quarter, and one quarter of 80 is 20.

#22 · Patterns and Relations

Solve for x: $x + 9 + 6 = 30$.

Answer

x equals 15. Combine the numbers on the left: $9 + 6 = 15$, so the equation is $x + 15 = 30$. Subtract 15 from both sides to get $x = 15$.

#23 · Geometry and Measurement

**A right triangle has legs of 3 cm and 4 cm.
How long is the hypotenuse?**

Answer

The hypotenuse is 5 cm. Use the Pythagorean theorem: leg squared plus leg squared = hypotenuse squared. $3^2 = 9$ and $4^2 = 16$, and $9 + 16 = 25$. The square root of 25 is 5.

#24 · Data Management and Probability

A fair six-sided die is rolled once. What is the probability of rolling a 4?

Answer

The probability is $\frac{1}{6}$. There is one favourable outcome (the 4) out of six equally likely outcomes. Probability = favourable outcomes divided by total possible outcomes.

#25 · Number Sense and Operations

Evaluate $6 + 2 \times 3$ using the order of operations.

Answer

The answer is 12. Following BEDMAS, do multiplication before addition: $2 \times 3 = 6$, then $6 + 6 = 12$. If you simply added left to right you would get a wrong answer of 24.

#26 · Patterns and Relations

Solve for x : $4x + 7 = 27$.

Answer

x equals 5. Subtract 7 from both sides to get $4x = 20$. Then divide both sides by 4 to get $x = 5$.

#27 · Geometry and Measurement

A box is 4 cm long, 3 cm wide, and 2 cm tall. What is its surface area?

Answer

The surface area is 52 square cm. The three face pairs have areas $4 \times 3 = 12$, $4 \times 2 = 8$, and $3 \times 2 = 6$. Add them ($12 + 8 + 6 = 26$) and double for both sides: $2 \times 26 = 52$.

#28 · Data Management and Probability

A bag has 3 red marbles and 7 blue marbles. What is the probability of picking a red marble?

Answer

The probability is $\frac{3}{10}$. There are 3 red marbles out of 10 total marbles ($3 + 7 = 10$). Probability = favourable outcomes divided by total outcomes.

#29 · Number Sense and Operations

Add the integers 7 and negative 12. What is the result?

Answer

The result is negative 5, written as -5. Starting at 7 and moving 12 steps down on a number line lands at -5. Adding a negative is the same as subtracting that value: $7 \text{ minus } 12 = -5$.

#30 · Patterns and Relations

The formula is $\text{distance} = \text{speed} \times \text{time}$. A trip of 120 km takes 4 hours. What is the speed?

Answer

The speed is 30 km per hour. Rearrange the formula to $\text{speed} = \text{distance} \text{ divided by time}$, so $\text{speed} = 120 \text{ divided by } 4 = 30$. Rearranging a formula means isolating the letter you want.

#31 · Geometry and Measurement

A rectangular prism is 5 cm long, 4 cm wide, and 3 cm tall. What is its volume?

Answer

The volume is 60 cubic cm. Multiply length by width by height: $5 \times 4 \times 3 = 60$. Volume measures how much space an object takes up.

#32 · Data Management and Probability

A coin is flipped twice. What is the probability of getting heads both times?

Answer

The probability is $\frac{1}{4}$. The flips are independent, so multiply the separate chances: $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$. Independent events do not affect each other.

#33 · Number Sense and Operations

A jacket costs \$60 and is on sale for 20% off. What is the sale price?

Answer

The sale price is \$48. The discount is 20% of \$60, or $0.20 \times 60 = \$12$. Subtract the discount from the original price: \$60 minus \$12 = \$48.

#34 · Patterns and Relations

One store sells 3 items for \$6 and another sells 5 of the same item for \$12. Which is the better buy?

Answer

The 3 for \$6 deal is better. Find the price per item: \$6 divided by 3 = \$2 each, while \$12 divided by 5 = \$2.40 each. The lower unit price is the better buy.

#35 · Geometry and Measurement

Convert 3.5 kilometres to metres.

Answer

The answer is 3500 metres. One kilometre equals 1000 metres, so multiply 3.5 by 1000: $3.5 \times 1000 = 3500$. Moving from a larger unit to a smaller one means multiplying.

#36 · Data Management and Probability

A fair six-sided die is rolled once. What is the probability of rolling a 1 or a 6?

Answer

The probability is $\frac{2}{6}$, which simplifies to $\frac{1}{3}$. These events are mutually exclusive because a single roll cannot be both 1 and 6, so add the separate chances: $\frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$.

#37 · Number Sense and Operations

A meal costs \$50 before tax. Sales tax is 13%. What is the total cost?

Answer

The total is \$56.50. The tax is 13% of \$50, or $0.13 \times 50 = \$6.50$. Add the tax to the original price: $\$50 + \$6.50 = \$56.50$.

#38 · Patterns and Relations

Solve for x: $\frac{2}{5} = \frac{x}{20}$.

Answer

x equals 8. Cross multiply: 5 times x = 2 times 20, so $5x = 40$. Divide both sides by 5 to get $x = 8$.

#39 · Geometry and Measurement

A photo is enlarged using a scale factor of 3. If the original width is 4 cm, what is the new width?

Answer

The new width is 12 cm. Multiply the original measurement by the scale factor: $4 \times 3 = 12$. A scale factor greater than 1 makes the shape larger, and every side is multiplied by the same factor.

#40 · Data Management and Probability

A circle graph is divided into 5 equal slices. What percent is one slice?

Answer

One slice is 20%. The whole circle graph represents 100%, and $100\% \div 5 = 20\%$. Each equal slice is one fifth of the whole.

#41 · Number Sense and Operations

What simple interest is earned on \$200 at 4% per year for 3 years?

Answer

The interest is \$24. Use interest = principal x rate x time: $200 \times 0.04 \times 3 = 24$. The principal is the starting amount, and the rate is written as a decimal.

#42 · Patterns and Relations

The pattern is 50, 42, 34, 26. What is the next term?

Answer

The next term is 18. The rule is subtract 8 each time: $26 \text{ minus } 8 = 18$. The pattern decreases by the same amount each step.

#43 · Geometry and Measurement

A cylinder has a radius of 3 cm and a height of 10 cm. Using pi as 3.14, what is its volume?

Answer

The volume is 282.6 cubic cm. Use volume = pi x radius squared x height: $3.14 \times 3 \times 3 \times 10 = 3.14 \times 90 = 282.6$. A cylinder's volume is the area of its circular end times its height.

#44 · Data Management and Probability

A bar graph shows four bars with values 5, 10, 15, and 20. Which value is on the tallest bar, and what is the total of all four?

Answer

The tallest bar is 20, and the total is 50. The tallest bar shows the greatest value. Add the four values to find the total: $5 + 10 + 15 + 20 = 50$.

#45 · Number Sense and Operations

A salesperson earns a 5% commission on a \$600 sale. How much commission is earned?

Answer

The commission is \$30. Multiply the sale amount by the commission percent written as a decimal: $0.05 \times 600 = 30$. Commission is a percent of the sale paid to the salesperson.

#46 · Patterns and Relations

Twice a number plus 7 equals 21. What is the number?

Answer

The number is 7. Set up the equation $2x + 7 = 21$. Subtract 7 from both sides to get $2x = 14$, then divide both sides by 2 to get $x = 7$. Use a letter like x to stand for the unknown number.

#47 · Geometry and Measurement

A recipe uses cups. Given that 1 cup = 250 mL, how many millilitres are in 3 cups?

Answer

The answer is 750 mL. Multiply the number of cups by the conversion rate: $3 \times 250 = 750$. Because the conversion is provided, multiply the starting amount by the rate.

#48 · Data Management and Probability

The numbers are 4, 7, 8, and 11. What is the median?

Answer

The median is 7.5. The list already has an even number of values, so find the two middle numbers, 7 and 8, and average them: $(7 + 8) \text{ divided by } 2 = 7.5$.

#49 · Number Sense and Operations

What do 2 to the power of 3 (2^3) and 5 squared (5^2) equal?

Answer

$2^3 = 8$ and $5^2 = 25$. The exponent tells you how many times to multiply the base by itself: $2 \times 2 \times 2 = 8$, and $5 \times 5 = 25$. Squaring means multiplying a number by itself once.

#50 · Patterns and Relations

The pattern is 1, 3, 5, 7. What is the next term?

Answer

The next term is 9. The rule is add 2 each time: $7 + 2 = 9$. This is a list of odd numbers, and each term is two more than the one before it.

#51 · Geometry and Measurement

What is the perimeter of a rectangle that is 6 m long and 4 m wide?

Answer

20 m. Perimeter is the total distance around the outside, so add all four sides: $6 + 4 + 6 + 4 = 20$. You can also use $2 \times (\text{length} + \text{width}) = 2 \times (6 + 4) = 20$.

#52 · Data Management and Probability

What is the mean of the numbers 4, 7, 8, and 9?

Answer

7. The mean is the average. Add the numbers: $4 + 7 + 8 + 9 = 28$, then divide by how many numbers there are, 4: $28 \div 4 = 7$.

#53 · Number Sense and Operations

What is $\frac{2}{7} + \frac{3}{7}$?

Answer

$\frac{5}{7}$. When the bottom numbers are the same, add only the top numbers: $2 + 3 = 5$, and keep the bottom number 7. The bottom number is how many equal parts, and it stays the same here.

#54 · Patterns and Relations

What is the next number in the pattern 4, 7, 10, 13, ...?

Answer

16. Each term goes up by 3, so $13 + 3 = 16$. In this arithmetic pattern you add the same amount at every step.

#55 · Geometry and Measurement

What is the area of a rectangle that is 6 cm long and 4 cm wide?

Answer

24 square cm. Area is the space inside a shape and is measured in square units. Multiply length $\times$ width:
 $6 \times 4 = 24$.

#56 · Data Management and Probability

What is the median of the numbers 3, 8, 5, 9, and 2?

Answer

5. The median is the middle value. Put the numbers in order: 2, 3, 5, 8, 9. With an odd count, the median is the one exactly in the middle.

#57 · Number Sense and Operations

Write 35% as a decimal.

Answer

0.35. Percent means out of 100, so divide 35 by 100: $35 \div 100 = 0.35$. You can also move the decimal point two places to the left.

#58 · Patterns and Relations

Write the ratio of 2 cats to 6 dogs in simplest form.

Answer

1 to 3, written 1:3. A ratio compares quantities of the same kind. Divide both parts by the greatest common factor, 2: $2 \div 2 = 1$ and $6 \div 2 = 3$.

#59 · Geometry and Measurement

What is the area of a triangle with a base of 8 m and a height of 5 m?

Answer

20 square m. A triangle is half of a rectangle with the same base and height. $\text{Area} = (\text{base} \times \text{height}) \div 2 = (8 \times 5) \div 2 = 40 \div 2 = 20$.

#60 · Data Management and Probability

What is the mode of the numbers 2, 5, 5, 7, and 9?

Answer

5. The mode is the value that appears most often. The number 5 shows up twice, more than any other.

#61 · Number Sense and Operations

What is $8 + 2 \times 3$?

Answer

14. Do the multiplication before the addition: $2 \times 3 = 6$, then $8 + 6 = 14$. BEDMAS means brackets, exponents, division and multiplication, then addition and subtraction.

#62 · Patterns and Relations

Solve for x: $\frac{3}{4} = \frac{9}{x}$.

Answer

$x = 12$. Cross multiply: $3 \times x = 4 \times 9$, so $3x = 36$. Then divide both sides by 3: $x = 36 \div 3 = 12$.

#63 · Geometry and Measurement

What is the circumference of a circle with a diameter of 10 cm? Use $\pi = 3.14$.

Answer

31.4 cm. Circumference is the distance around the circle. Multiply π by the diameter: $3.14 \times 10 = 31.4$.

#64 · Data Management and Probability

What is the range of the numbers 3, 9, 6, and 1?

Answer

8. The range is the spread from lowest to highest. Subtract the smallest from the biggest: $9 - 1 = 8$.

#65 · Number Sense and Operations

What is 15% of \$80?

Answer

\$12. Change 15% into 0.15 and multiply: $0.15 \times 80 = 12$. You could also find 10% (\$8), then add 5% (\$4) to get \$12.

#66 · Patterns and Relations

A 5 kg bag of rice costs \$15. What is the unit price per kilogram?

Answer

\$3 per kg. A unit rate is the cost for one unit. Divide the total cost by the number of kilograms: $15 \div 5 = 3$.

#67 · Geometry and Measurement

What is the area of a circle with a radius of 3 cm? Use $\pi = 3.14$.

Answer

28.26 square cm. Area is the space inside the circle. Multiply π by the radius squared: $3.14 \times 3 \times 3 = 3.14 \times 9 = 28.26$.

#68 · Data Management and Probability

Most employees earn near \$40,000, but one person earns \$120,000. Which average is pulled upward, the mean or the median?

Answer

The mean. That \$120,000 is an outlier, a value far from the rest, and it raises the average a lot. The median stays near the middle because it only depends on the middle value.

#69 · Number Sense and Operations

**A jacket costs \$40 and sales tax is 13%.
What is the total price?**

Answer

\$45.20. Find the tax first: $0.13 \times 40 = \$5.20$. Then add it to the price: $40 + 5.20 = 45.20$.

#70 · Patterns and Relations

**A car travels 240 km in 3 hours. What is its
average speed?**

Answer

80 km/h. Average speed is the distance travelled in one hour, so use distance $\div$ time: $240 \div 3 = 80$.

#71 · Geometry and Measurement

**A right triangle has legs of 3 cm and 4 cm.
How long is the hypotenuse?**

Answer

5 cm. The hypotenuse is the side opposite the right angle. Use $a^2 + b^2 = c^2$: $9 + 16 = 25$, and the square root of 25 is 5.

#72 · Data Management and Probability

In a bar graph, what does the height of each bar tell you?

Answer

It tells the value or amount for that category. Compare the bar heights to see which category has the most or the least. A bar graph is used to compare amounts across different groups.

#73 · Number Sense and Operations

A \$120 bike is marked 25% off. What is the sale price?

Answer

\$90. The discount is $0.25 \times 120 = \$30$. Subtract that from the original price: $120 - 30 = 90$.

#74 · Patterns and Relations

A map scale says 1 cm = 5 km. Two towns are 6 cm apart on the map. How far apart are they in real life?

Answer

30 km. A scale shows the link between the drawing and real life. Multiply the map distance by the scale: $6 \times 5 = 30$.

#75 · Geometry and Measurement

What is the volume of a box that is 5 cm long, 4 cm wide, and 3 cm tall?

Answer

60 cubic cm. Volume is the space inside a 3D shape and is measured in cubic units. Multiply length $\times$ width $\times$ height: $5 \times 4 \times 3 = 60$.

#76 · Data Management and Probability

A line graph shows a temperature going from 10 up to 20 degrees over 3 hours. What does the rising line show?

Answer

The temperature increased over time. A line graph shows how a value changes across a period, and a line moving upward means the value is getting larger.

#77 · Number Sense and Operations

Write 2 and 3/4 as an improper fraction.

Answer

11/4. Multiply the whole number by the bottom number: $2 \times 4 = 8$. Add the top number: $8 + 3 = 11$. Keep the bottom number 4, so the answer is 11/4.

#78 · Patterns and Relations

The perimeter of a rectangle is $P = 2 \times (\text{length} + \text{width})$. If length is 7 and width is 3, what is P?

Answer

20. To solve a formula, replace each letter with its number: $P = 2 \times (7 + 3) = 2 \times 10 = 20$. Do the part in brackets first.

#79 · Geometry and Measurement

What is the volume of a cylinder with a radius of 2 cm and a height of 10 cm? Use $\pi = 3.14$.

Answer

125.6 cubic cm. Volume of a cylinder = $\pi \times \text{radius squared} \times \text{height} = 3.14 \times 2 \times 2 \times 10 = 3.14 \times 4 \times 10 = 3.14 \times 40 = 125.6$.

#80 · Data Management and Probability

A circle graph is split into slices that add to 100%. One slice is 50%. How much of the whole is that slice?

Answer

Half. The whole circle is 100%, so a 50% slice covers half of it. All the slices together always add to 100%.

#81 · Number Sense and Operations

What is $3.25 + 1.4$?

Answer

4.65. Line up the decimal points, writing 1.4 as 1.40, then add: $3.25 + 1.40 = 4.65$. Always line up the decimal points before adding or subtracting decimals.

#82 · Patterns and Relations

**A recipe uses 2 cups of flour for 12 muffins.
How many cups are needed for 30 muffins?**

Answer

5 cups. Set up a proportion: $2/12 = x/30$. Cross multiply: $12x = 60$, then divide: $x = 5$. Keep the same units on top and on the bottom.

#83 · Geometry and Measurement

What is the surface area of a box that is 2 m long, 3 m wide, and 4 m tall?

Answer

52 square m. Surface area is the total area of all six faces. Add the three different face areas and double: $2 \times (2 \times 3 + 3 \times 4 + 2 \times 4) = 2 \times (6 + 12 + 8) = 2 \times 26 = 52$.

#84 · Data Management and Probability

A scatterplot shows points that rise as you move to the right and up. What does that pattern show?

Answer

A positive relationship. As one value increases, the other value tends to increase too. A scatterplot shows whether two variables are related and in which direction.

#85 · Number Sense and Operations

\$1,000 is invested at 5% per year for 2 years. What is the total interest earned?

Answer

\$100. Interest = principal $\times$ rate $\times$ time = $1,000 \times 0.05 \times 2 = 100$. The rate must be written as a decimal, so 5% becomes 0.05.

#86 · Patterns and Relations

The pattern 20, 16, 12, 8 goes down each time. What is the rule?

Answer

Subtract 4 each time: $20 - 4 = 16$, $16 - 4 = 12$, $12 - 4 = 8$. To find the rule, check the change from one term to the next.

#87 · Geometry and Measurement

A shape is enlarged by a scale factor of 3. A side that was 4 cm long is now how long?

Answer

12 cm. Multiply the original length by the scale factor: $4 \times 3 = 12$. A scale factor greater than 1 makes the figure larger, and a scale factor less than 1 makes it smaller.

#88 · Data Management and Probability

A bag has 3 red and 5 blue marbles. What is the probability of picking a red marble?

Answer

3 out of 8, written $\frac{3}{8}$. Probability is favourable outcomes $\div$ total outcomes. The total is $3 + 5 = 8$, and there are 3 red marbles.

#89 · Number Sense and Operations

What is $-4 + 9$?

Answer

5. Start at -4 and move 9 steps toward the positive numbers, ending at 5. Because 9 is larger than 4, the answer is positive: $9 - 4 = 5$.

#90 · Patterns and Relations

If 3 books cost \$18, how much do 6 books cost?

Answer

\$36. Double the books and the cost together: 3 to 6 is times 2, so $18 \times 2 = 36$. Or set up $3/18 = 6/x$ and cross multiply to get $x = 36$.

#91 · Geometry and Measurement

How many centimetres are in 2.5 metres?

Answer

250 cm. There are 100 cm in 1 m, so multiply: $2.5 \times 100 = 250$. Going from a larger unit to a smaller unit means multiplying.

#92 · Data Management and Probability

You flip a coin and roll a six-sided die. What is the probability of getting heads and a 6?

Answer

1 out of 12, written $1/12$. For independent events multiply the chances: $1/2 \times 1/6 = 1/12$. The coin flip does not change the die roll.

#93 · Number Sense and Operations

Solve for x : $x + 7 = 15$.

Answer

$x = 8$. Subtract 7 from both sides to keep the equation balanced: $x = 15 - 7 = 8$.

#94 · Patterns and Relations

Share \$50 between two people in the ratio 2:3. What is the smaller share?

Answer

\$20. Add the ratio parts: $2 + 3 = 5$. One part is $50 \div 5 = \$10$. The smaller share is 2 parts, so $2 \times 10 = \$20$.

#95 · Geometry and Measurement

An L-shaped room is a 4 m by 4 m square with a 2 m by 3 m rectangle attached. What is the total area?

Answer

22 square m. Break a composite shape into simple shapes, find each area, and add: $4 \times 4 = 16$ and $2 \times 3 = 6$, so $16 + 6 = 22$.

#96 · Data Management and Probability

Events A and B cannot happen at the same time. If the chance of A is 0.3 and the chance of B is 0.5, what is the chance that A or B happens?

Answer

0.8. Mutually exclusive events cannot overlap, so you add their probabilities: $0.3 + 0.5 = 0.8$.

#97 · Number Sense and Operations

Solve for x : $2x + 3 = 11$.

Answer

$x = 4$. Subtract 3 from both sides: $2x = 8$. Then divide both sides by 2: $x = 4$.

#98 · Patterns and Relations

A worker earns \$18 per hour. How much is earned for 7 hours of work?

Answer

\$126. A rate is an amount for one unit, so multiply the rate by the number of hours: $18 \times 7 = 126$.

#99 · Geometry and Measurement

How many litres are in 2,500 millilitres?

Answer

2.5 L. There are 1,000 mL in 1 L, so divide: $2,500 \div 1,000 = 2.5$. Going from a smaller unit to a larger unit means dividing.

#100 · Data Management and Probability

A table shows 12 students chose apples and 8 chose bananas. How many students chose fruit in total?

Answer

20. Read the two numbers from the table and add them: $12 + 8 = 20$. A table organizes data in rows and columns so it is easy to read.

#101 · Number Sense and Operations

How do you write the decimal 0.75 as a fraction in simplest form?

Answer

3 over 4. The decimal 0.75 means 75 out of 100, so write $\frac{75}{100}$ and then divide the top and bottom by 25. 75 divided by 25 is 3, and 100 divided by 25 is 4.

#102 · Patterns and Relations

What is the next number in the pattern 3, 7, 11, 15, ...?

Answer

19. The pattern adds 4 each time: 3 plus 4 is 7, 7 plus 4 is 11, and 11 plus 4 is 15. So the next term is 15 plus 4, which is 19.

#103 · Geometry and Measurement

What is the perimeter of a rectangle with length 8 m and width 5 m?

Answer

26 m. Perimeter is the distance around the shape. Add the length and the width, then double that total: 2 times (8 plus 5) equals 2 times 13, which is 26.

#104 · Data Management and Probability

What is the mean (average) of the numbers 4, 8, 12, and 16?

Answer

10. Add all four numbers to get 40, then divide by 4 because there are 4 values. 40 divided by 4 is 10.

#105 · Number Sense and Operations

What is $\frac{1}{2}$ times $\frac{3}{4}$?

Answer

$\frac{3}{8}$ over 8. To multiply fractions, multiply the top numbers and multiply the bottom numbers: 1 times 3 is 3, and 2 times 4 is 8.

#106 · Patterns and Relations

What is the value of x in the equation x plus 9 equals 14?

Answer

5. To undo the plus 9, subtract 9 from both sides. x equals 14 minus 9, which is 5.

#107 · Geometry and Measurement

What is the area of a rectangle with length 6 cm and width 4 cm?

Answer

24 square cm. Area is the space inside the shape. Multiply length times width: 6 times 4 equals 24.

#108 · Data Management and Probability

What is the median of the data set 5, 2, 9, 4, and 7?

Answer

5. The median is the middle value. First put the numbers in order: 2, 4, 5, 7, 9. The middle value is 5.

#109 · Number Sense and Operations

How do you write 12 percent as a decimal?

Answer

0.12. Percent means out of 100, so 12 percent is 12 out of 100, written as $12/100$. Divide 12 by 100 to move the decimal two places left, which gives 0.12.

#110 · Patterns and Relations

What is the value of x in the equation $3x$ plus 4 equals 19?

Answer

5. Subtract 4 from both sides to get $3x$ equals 15, then divide both sides by 3. x equals 15 divided by 3, which is 5.

#111 · Geometry and Measurement

What is the area of a triangle with base 10 cm and height 6 cm?

Answer

30 square cm. Area of a triangle is one half of base times height. Multiply 10 times 6 to get 60, then divide by 2 to get 30.

#112 · Data Management and Probability

What is the mode of the data set 3, 7, 3, 9, 7, and 3?

Answer

3. The mode is the value that appears most often. The number 3 appears three times, which is more often than any other number.

#113 · Number Sense and Operations

What is 2 and $\frac{1}{4}$ plus 1 and $\frac{1}{2}$?

Answer

3 and $\frac{3}{4}$. Change each mixed number to a fraction: 2 and $\frac{1}{4}$ is $\frac{9}{4}$, and 1 and $\frac{1}{2}$ is $\frac{6}{4}$. Add the tops to get $\frac{15}{4}$, which is the same as 3 and $\frac{3}{4}$.

#114 · Patterns and Relations

What is the next number in the pattern 2, 4, 8, 16, ...?

Answer

32. Each term is double the one before it: 2 times 2 is 4, 4 times 2 is 8, and 8 times 2 is 16. So the next term is 16 times 2, which is 32.

#115 · Geometry and Measurement

A right triangle has legs of 3 and 4. What is the length of the hypotenuse?

Answer

5. Use the Pythagorean theorem: leg squared plus leg squared equals hypotenuse squared. 3 squared is 9, and 4 squared is 16. Add them to get 25, and the square root of 25 is 5.

#116 · Data Management and Probability

What is the range of the data set 12, 7, 20, and 5?

Answer

15. The range is the largest value minus the smallest value. The largest is 20 and the smallest is 5, so 20 minus 5 is 15.

#117 · Number Sense and Operations

What is 15 percent of 80?

Answer

12. Write 15 percent as the decimal 0.15, then multiply by 80. 0.15 times 80 equals 12.

#118 · Patterns and Relations

A formula says d equals $8t$. What is the value of d when t is 3?

Answer

24. To use a formula, put the given number in place of the letter. d equals 8 times t , so d equals 8 times 3, which is 24.

#119 · Geometry and Measurement

What is the volume of a box that is 5 m long, 3 m wide, and 2 m tall?

Answer

30 cubic m. Volume is length times width times height. Multiply 5 times 3 to get 15, then multiply by 2 to get 30.

#120 · Data Management and Probability

How does adding one very high number to a data set affect the mean?

Answer

It raises the mean. The mean is the total divided by the count. A very large new value increases the total a lot, so the mean moves up toward that high number.

#121 · Number Sense and Operations

What is the value of 2 plus 3 times 4?

Answer

14. Follow the order of operations. Do the multiplication first: 3 times 4 is 12, then add 2 to get 14. Doing 2 plus 3 first would give the wrong answer of 20.

#122 · Patterns and Relations

What is the value of x in the equation $5x$ minus 2 equals 13?

Answer

3. Add 2 to both sides to get $5x$ equals 15, then divide both sides by 5. x equals 15 divided by 5, which is 3.

#123 · Geometry and Measurement

What is the circumference of a circle with radius 7 cm? Use pi as $\frac{22}{7}$.

Answer

44 cm. Circumference is 2 times pi times the radius. Using $\frac{22}{7}$ for pi, the working is 2 times $\frac{22}{7}$ times 7. The 7s cancel, leaving 2 times 22, which is 44.

#124 · Data Management and Probability

A bag has 4 red marbles and 6 blue marbles. What is the probability of drawing a red marble?

Answer

2 out of 5, or $\frac{2}{5}$. Probability is the number of good outcomes divided by the total outcomes. There are 4 red marbles and 10 total, so $\frac{4}{10}$, which simplifies to $\frac{2}{5}$.

#125 · Number Sense and Operations

How do you write the fraction $\frac{7}{10}$ as a decimal?

Answer

0.7. A fraction means to divide the top number by the bottom number. Divide 7 by 10, which gives 0.7 because the decimal point moves one place left.

#126 · Patterns and Relations

A number plus 8 equals 20. Write an equation for this sentence and find the number.

Answer

x plus 8 equals 20, and x is 12. Let x stand for the unknown number. To solve, subtract 8 from both sides: x equals 20 minus 8, which is 12.

#127 · Geometry and Measurement

What is the area of a circle with radius 5 cm? Use pi as 3.14.

Answer

78.5 square cm. Area of a circle is pi times the radius squared. The radius squared is 5 times 5, which is 25. Multiply 3.14 by 25 to get 78.5.

#128 · Data Management and Probability

A coin is flipped and a die is rolled. What is the probability of getting a head and a 6?

Answer

1 out of 12. For two independent events, multiply the probabilities. The chance of a head is $\frac{1}{2}$ and the chance of a 6 is $\frac{1}{6}$. Multiply $\frac{1}{2}$ times $\frac{1}{6}$ to get $\frac{1}{12}$.

#129 · Number Sense and Operations

How do you write the improper fraction $11/4$ as a mixed number?

Answer

2 and $3/4$. Divide the top by the bottom: 11 divided by 4 gives 2 with a remainder of 3. The whole number is 2 and the remainder sits over the 4, giving 2 and $3/4$.

#130 · Patterns and Relations

What is the next number in the pattern 20, 17, 14, 11, ...?

Answer

8. The pattern subtracts 3 each time: 20 minus 3 is 17, 17 minus 3 is 14, and 14 minus 3 is 11. So the next term is 11 minus 3, which is 8.

#131 · Geometry and Measurement

How many millimeters are in 3.5 centimeters?

Answer

35 mm. One centimeter equals 10 millimeters, so multiply by 10. 3.5 times 10 equals 35.

#132 · Data Management and Probability

A die is rolled. What is the probability of rolling a 2 or a 4?

Answer

1 out of 3, or $\frac{1}{3}$. These two results cannot happen at the same time, so add their probabilities. Each is $\frac{1}{6}$, so $\frac{1}{6}$ plus $\frac{1}{6}$ is $\frac{2}{6}$, which simplifies to $\frac{1}{3}$.

#133 · Number Sense and Operations

What is negative 5 plus 9?

Answer

4. Start at negative 5 and move up 9 places toward the positive side. Negative 5 plus 5 reaches 0, and the remaining 4 brings you to 4.

#134 · Patterns and Relations

What is the value of x in the equation x divided by 3 equals 12?

Answer

36. To undo dividing by 3, multiply both sides by 3. x equals 12 times 3, which is 36.

#135 • Geometry and Measurement

A square has a side length of 4 cm. If every side is doubled, what is the new area?

Answer

64 square cm. Doubling 4 cm gives a side of 8 cm. Area of a square is side times side, so 8 times 8 equals 64.

#136 • Data Management and Probability

On a bar graph, the horizontal axis lists fruits and the vertical axis shows the count. The Apple bar reaches 25. How many apples were counted?

Answer

25 apples. The height of a bar tells you the value for that category. Since the Apple bar reaches 25 on the count axis, there are 25 apples.

#137 · Number Sense and Operations

Round the decimal 3.67 to the nearest tenth.

Answer

3.7. The tenths digit is 6. Look at the digit after it, which is 7. Since 7 is 5 or more, round the tenths digit up from 6 to 7, giving 3.7.

#138 · Patterns and Relations

A formula says C equals 5 plus $2n$. What is the value of C when n is 6?

Answer

17. Replace n with 6: C equals 5 plus 2 times 6. Multiply 2 times 6 to get 12, then add 5 to get 17.

#139 · Geometry and Measurement

What is the volume of a cube with an edge length of 3 cm?

Answer

27 cubic cm. For a cube, multiply the edge by itself three times: 3 times 3 is 9, and 9 times 3 is 27.

#140 · Data Management and Probability

On a line graph, the temperature line goes steadily upward from left to right. What does this tell you?

Answer

The temperature is increasing over time. A rising line on a line graph shows that the value on the vertical axis grows higher as the value on the horizontal axis moves forward.

#141 · Number Sense and Operations

How do you simplify the ratio 12 to 18?

Answer

2 to 3. Divide both parts of the ratio by the same number. Both 12 and 18 divide evenly by 6, so 12 divided by 6 is 2, and 18 divided by 6 is 3.

#142 · Patterns and Relations

What is the next number in the pattern 1, 2, 4, 7, 11, ...?

Answer

16. The gaps between the terms grow by 1: the steps are plus 1, plus 2, plus 3, and plus 4. The next step is plus 5, so 11 plus 5 is 16.

#143 · Geometry and Measurement

Two angles of a triangle are 50 degrees and 60 degrees. What is the third angle?

Answer

70 degrees. The three angles of any triangle always add to 180 degrees. Add 50 and 60 to get 110, then subtract from 180 to get 70.

#144 · Data Management and Probability

The probability that it will rain is 0.4. What is the probability that it will not rain?

Answer

0.6. The chance of rain plus the chance of no rain must add to 1. Subtract 0.4 from 1 to get 0.6.

#145 · Number Sense and Operations

A car travels 240 km in 3 hours. What is its average speed in km per hour?

Answer

80 km per hour. Speed equals distance divided by time. Divide 240 by 3 to get 80.

#146 · Patterns and Relations

What is the value of x in the equation x minus 7 equals 12?

Answer

19. To undo the minus 7, add 7 to both sides. x equals 12 plus 7, which is 19.

#147 · Geometry and Measurement

What is the volume of a cylinder with radius 3 cm and height 4 cm? Use pi as 3.14.

Answer

113.04 cubic cm. Volume of a cylinder is pi times radius squared times height. The radius squared is 3 times 3, which is 9. Multiply 3.14 by 9 to get 28.26, then multiply by 4 to get 113.04.

#148 · Data Management and Probability

A circle graph shows that 25 percent of a day is spent on sports. What fraction of the day is that?

Answer

1 out of 4. Percent means out of 100, so 25 percent is 25/100. Divide the top and bottom by 25 to get 1/4.

#149 · Number Sense and Operations

What is the simple interest on 500 dollars at 4 percent per year for 2 years?

Answer

40 dollars. Simple interest is principal times rate times time. The rate 4 percent is 0.04 as a decimal. Multiply 500 times 0.04 to get 20, then multiply by 2 years to get 40.

#150 · Patterns and Relations

Write an equation for the sentence: twice a number plus 5 is 13. Then find the number.

Answer

2x plus 5 equals 13, and x is 4. Let x be the unknown number. Subtract 5 from both sides to get 2x equals 8, then divide by 2. x equals 8 divided by 2, which is 4.

#151 · Geometry and Measurement

**A rectangle is 6 cm long and 4 cm wide.
What is its perimeter?**

Answer

The perimeter is 20 cm. Add all four sides: $6 + 4 + 6 + 4 = 20$ cm. Or use $P = 2(l + w) = 2(6 + 4) = 2(10) = 20$ cm. Perimeter is the distance around a shape.

#152 · Data Management and Probability

**What is the mean (average) of the numbers
4, 8, 10, and 14?**

Answer

The mean is 9. Add all the numbers: $4 + 8 + 10 + 14 = 36$. Then divide by how many numbers there are: $36 / 4 = 9$.

#153 · Number Sense and Operations

What is $\frac{3}{4}$ written as a decimal and as a percent?

Answer

$\frac{3}{4}$ as a decimal is 0.75 and as a percent is 75%.
Divide the top number by the bottom number: $3 \div 4 = 0.75$. To change a decimal to a percent, move the decimal point two places to the right, so 0.75 becomes 75%.

#154 · Patterns and Relations

What is the next number in the pattern 2, 5, 8, 11, ...?

Answer

The next number is 14. Each number goes up by 3.
Add 3 to the last number: $11 + 3 = 14$. The pattern is add 3 each time: 2, 5, 8, 11, 14.

#155 · Geometry and Measurement

A triangle has a base of 10 cm and a height of 6 cm. What is its area?

Answer

The area is 30 cm². Use $A = \frac{1}{2} \times \text{base} \times \text{height} = 0.5 \times 10 \times 6 = 30$. A triangle takes up half as much space as a rectangle with the same base and height.

#156 · Data Management and Probability

What is the median of the numbers 3, 7, 2, 9, and 5?

Answer

The median is 5. Put the numbers in order: 2, 3, 5, 7, 9. The median is the middle value, which is 5. With an odd amount of numbers, it is the one in the centre.

#157 · Number Sense and Operations

What is $\frac{1}{4} + \frac{1}{8}$? Give your answer as a fraction in lowest terms.

Answer

$\frac{1}{4} + \frac{1}{8} = \frac{3}{8}$. Find a common denominator, which is a bottom number both fractions can use. The smallest number 4 and 8 both divide into is 8. Change $\frac{1}{4}$ to $\frac{2}{8}$, then add $\frac{2}{8} + \frac{1}{8} = \frac{3}{8}$.

#158 · Patterns and Relations

Solve for x: $x + 7 = 12$.

Answer

$x = 5$. Subtract 7 from both sides to get x alone: $x = 12 - 7 = 5$. Whatever you do to one side of an equation, you must do to the other side.

#159 · Geometry and Measurement

**A box is 5 cm long, 3 cm wide, and 4 cm tall.
What is its volume?**

Answer

The volume is 60 cm³. Use $V = \text{length} \times \text{width} \times \text{height} = 5 \times 3 \times 4 = 60 \text{ cm}^3$. Volume is the space inside a 3D shape, measured in cubic units.

#160 · Data Management and Probability

What is the mode of the numbers 2, 5, 5, 8, 5, 9?

Answer

The mode is 5. The mode is the number that appears most often. In this list, 5 shows up three times, more than any other number.

#161 · Number Sense and Operations

A jacket costs \$80 and is on sale for 25% off. What is the sale price?

Answer

The sale price is \$60. Find 25% of \$80: $0.25 \times 80 = \$20$ off. Subtract that from the original price: $80 - 20 = \$60$. Or multiply directly: $80 \times 0.75 = \$60$.

#162 · Patterns and Relations

Solve for x: $3x = 21$.

Answer

$x = 7$. Divide both sides by 3: $x = 21 / 3 = 7$. The expression $3x$ means 3 times x , so the opposite of multiplying by 3 is dividing by 3.

#163 · Geometry and Measurement

A right triangle has legs of 3 cm and 4 cm. What is the length of the hypotenuse, the longest side?

Answer

The hypotenuse is 5 cm. Use $a^2 + b^2 = c^2$: $3^2 + 4^2 = 9 + 16 = 25 = c^2$, so $c = \text{the square root of } 25 = 5$ cm. The hypotenuse is the side across from the right angle.

#164 · Data Management and Probability

What is the range of the numbers 12, 7, 20, 15, and 3?

Answer

The range is 17. Subtract the smallest number from the largest number: $20 - 3 = 17$. The range shows how spread out the data is.

#165 · Number Sense and Operations

What is $(-6) \times (-4)$?

Answer

The answer is 24. When you multiply two negative numbers, the result is positive. Multiply the numbers and ignore the signs: $6 \times 4 = 24$, then leave it positive.

#166 · Patterns and Relations

**The perimeter of a rectangle is $P = 2(l + w)$.
What is the perimeter when the length is 8 cm and the width is 5 cm?**

Answer

The perimeter is 26 cm. Put the values into the formula: $P = 2(8 + 5) = 2(13) = 26$ cm. Add the length and width first, then multiply by 2.

#167 · Geometry and Measurement

A circle has a diameter of 10 cm. What is its circumference? Use $\pi = 3.14$.

Answer

The circumference is 31.4 cm. Use $C = \pi \times \text{diameter} = 3.14 \times 10 = 31.4$ cm. Circumference is the distance around a circle, and π is about 3.14.

#168 · Data Management and Probability

The ages are 10, 11, 11, 12, and 60. Does the outlier 60 pull the mean up or down, and is the median a better measure here?

Answer

The outlier pulls the mean up, so the median is the better measure. The mean is $10 + 11 + 11 + 12 + 60 = 104$, divided by 5 = 20.8, but the median is 11. One very large number makes the mean far from the typical age.

#169 · Number Sense and Operations

**A meal costs \$40 and the sales tax is 13%.
What is the total cost including tax?**

Answer

The total is \$45.20. Find 13% of \$40: $0.13 \times 40 = \$5.20$. Add that to the original price: $40 + 5.20 = \$45.20$. Or multiply: $40 \times 1.13 = \$45.20$.

#170 · Patterns and Relations

**A taxi charges a \$3 flat fee plus \$2 per
kilometre. How far can you travel for \$15?**

Answer

You can travel 6 km. The equation is $2x + 3 = 15$. Subtract 3 from both sides: $2x = 12$, then divide both sides by 2: $x = 6$ km.

#171 · Geometry and Measurement

A circle has a radius of 5 cm. What is its area? Use $\pi = 3.14$.

Answer

The area is 78.5 cm². Use $A = \pi \times \text{radius squared} = 3.14 \times 5 \text{ squared} = 3.14 \times 25 = 78.5 \text{ cm}^2$. The radius is half the diameter, and radius squared means the radius times itself.

#172 · Data Management and Probability

A bag has 3 red marbles and 7 blue marbles. What is the probability of drawing a red marble?

Answer

The probability is $\frac{3}{10}$, or 0.3, or 30%. There are 3 red marbles out of 10 marbles in total. Probability = favourable outcomes divided by total outcomes = $\frac{3}{10} = 0.3$.

#173 · Number Sense and Operations

Using order of operations, what is $3 + 4 \times 2$?

Answer

The answer is 11. Multiplication comes before addition, so do $4 \times 2 = 8$ first, then add 3: $3 + 8 = 11$. If you added first you would get 14, which is wrong.

#174 · Patterns and Relations

A pattern of squares starts with 1, 4, 9, 16. What is the next number?

Answer

The next number is 25. These are square numbers: 1×1 , 2×2 , 3×3 , 4×4 . The next one is $5 \times 5 = 25$. The gaps between terms go 3, 5, 7, then 9.

#175 · Geometry and Measurement

A cube has a side length of 3 cm. What is its surface area?

Answer

The surface area is 54 cm². A cube has 6 identical square faces. Each face has area $3 \times 3 = 9$ cm², so the total is $6 \times 9 = 54$ cm². Surface area is the total area of all the outside faces.

#176 · Data Management and Probability

You flip a coin and roll a die. What is the probability of getting heads AND a 4?

Answer

The probability is $\frac{1}{12}$. The chance of heads is $\frac{1}{2}$ and the chance of rolling a 4 is $\frac{1}{6}$. For independent events, where one does not change the other, multiply: $\frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$.

#177 · Number Sense and Operations

A restaurant bill is \$60 and you leave a 15% tip. How much is the tip?

Answer

The tip is \$9. Multiply $0.15 \times 60 = \$9$. For a quick estimate, 10% of \$60 is \$6 and 5% is \$3, so 15% is $\$6 + \$3 = \$9$.

#178 · Patterns and Relations

The distance formula is $d = r \times t$. How far do you travel at 60 km/h for 3 hours?

Answer

You travel 180 km. Put the values into the formula:
 $d = 60 \times 3 = 180$ km. Rate times time gives distance, so multiply the speed by the hours travelled.

#179 · Geometry and Measurement

How many centimetres are in 2.5 metres?

Answer

There are 250 centimetres. There are 100 centimetres in 1 metre, so multiply: $2.5 \times 100 = 250$ cm. To change metres to centimetres, move the decimal point two places to the right.

#180 · Data Management and Probability

A spinner lands on red with probability 0.4, on blue with 0.35, and on green with 0.25. What is the probability of landing on red or blue?

Answer

The probability is 0.75. These are mutually exclusive events, meaning the spinner can only land on one colour at a time. For mutually exclusive events, add the probabilities: $0.4 + 0.35 = 0.75$.

#181 · Number Sense and Operations

What is $2\frac{3}{4}$ written as an improper fraction, where the top number is bigger than the bottom?

Answer

$2\frac{3}{4} = \frac{11}{4}$. Multiply the whole number by the bottom number: $2 \times 4 = 8$. Add the top number: $8 + 3 = 11$. Keep the same bottom number: $\frac{11}{4}$.

#182 · Patterns and Relations

Solve for x: $2x + 5 = 17$.

Answer

$x = 6$. Subtract 5 from both sides: $2x = 12$, then divide both sides by 2: $x = 6$. Check it: $2(6) + 5 = 17$.

#183 • Geometry and Measurement

A photo is 4 cm wide and 6 cm tall. It is enlarged so the new width is 12 cm. What is the new height?

Answer

The new height is 18 cm. The scale factor is $12 / 4 = 3$, so the photo is tripled. Multiply the old height by 3: $6 \times 3 = 18$ cm. The width and height grow by the same factor.

#184 • Data Management and Probability

A bar graph shows that 20 students like apples, 15 like bananas, and 10 like oranges. How many students were surveyed in total?

Answer

A total of 45 students were surveyed. Add the heights of the bars: $20 + 15 + 10 = 45$. A bar graph shows the count for each category, and the total is the sum of all the bars.

#185 · Number Sense and Operations

If 5 pens cost \$7.50, how much do 8 pens cost?

Answer

8 pens cost \$12.00. First find the unit price: $7.50 / 5 = \$1.50$ per pen. Multiply by 8: $1.50 \times 8 = \$12.00$. Or use a proportion: $5/7.50 = 8/x$, so $x = 8 \times 7.50 / 5 = 12$.

#186 · Patterns and Relations

Solve for x: x divided by 4 = 9.

Answer

$x = 36$. Multiply both sides by 4: $x = 9 \times 4 = 36$. Because x is divided by 4, the opposite operation is multiplication.

#187 · Geometry and Measurement

An L-shaped room is made of two rectangles: one is 4 m by 3 m and the other is 2 m by 3 m. What is the total area?

Answer

The total area is 18 m². Add the areas of the two rectangles: $(4 \times 3) + (2 \times 3) = 12 + 6 = 18$ m². For a composite shape, split it into simpler shapes, find each area, then add them.

#188 · Data Management and Probability

In a circle graph, one section is labelled 25%. What fraction of the whole does this section represent?

Answer

It represents $\frac{1}{4}$. A circle graph shows parts of a whole that add up to 100%. Change 25% to a fraction: $\frac{25}{100}$, which reduces to $\frac{1}{4}$.

#189 · Number Sense and Operations

What is 30% of 150?

Answer

30% of 150 is 45. Change 30% to a decimal: 0.30, then multiply: $0.30 \times 150 = 45$. Or take 10% of 150, which is 15, then multiply by 3: $15 \times 3 = 45$.

#190 · Patterns and Relations

A phone plan costs \$25 per month plus \$0.10 per minute. What is the bill for 120 minutes in one month?

Answer

The bill is \$37. Use the expression $25 + (0.10 \times 120)$. Do the multiplication first: $0.10 \times 120 = \$12$, then add: $25 + 12 = \$37$.

#191 · Geometry and Measurement

A parallelogram has a base of 8 cm and a height of 5 cm. What is its area?

Answer

The area is 40 cm². Use $A = \text{base} \times \text{height} = 8 \times 5 = 40 \text{ cm}^2$. The height is the straight up distance from the base to the top, not the length of the slanted side.

#192 · Data Management and Probability

A scatterplot shows that more study time is paired with higher test scores. What does this pattern show?

Answer

It shows a positive trend: as study time goes up, test scores tend to go up. In a scatterplot, points that rise from left to right show a positive relationship between the two things being measured.

#193 · Number Sense and Operations

A salesperson earns an 8% commission on sales. What is the commission on \$2,500 in sales?

Answer

The commission is \$200. Change 8% to a decimal: 0.08, then multiply by the sales total: $0.08 \times 2500 = \$200$. Commission is a percent of the sales amount.

#194 · Patterns and Relations

The area of a rectangle is $A = l \times w$. What is the area when the length is 9 cm and the width is 4 cm?

Answer

The area is 36 cm². Put the values into the formula: $A = 9 \times 4 = 36$. Area is the space inside a shape, measured in square units.

#195 · Geometry and Measurement

A cylinder has a radius of 2 cm and a height of 5 cm. What is its volume? Use $\pi = 3.14$.

Answer

The volume is 62.8 cm³. Use $V = \pi \times \text{radius squared} \times \text{height} = 3.14 \times 2^2 \times 5 = 3.14 \times 4 \times 5 = 62.8 \text{ cm}^3$. First find the area of the circular base, then multiply by the height.

#196 · Data Management and Probability

A number puzzle says: take a number, double it, then add 6. You get 20. What was the starting number?

Answer

The starting number is 7. Work backwards. Undo the add 6 with $20 - 6 = 14$, then undo the doubling with $14 \div 2 = 7$. Do the opposite operations in reverse order.

#197 · Number Sense and Operations

You put \$1,000 in a savings account that pays 5% simple interest per year. How much interest do you earn in 2 years?

Answer

You earn \$100 in interest. Use $\text{interest} = \text{principal} \times \text{rate} \times \text{time}$, so $1000 \times 0.05 \times 2 = \100 . The principal is the starting money, the rate is 0.05, and the time is 2 years.

#198 · Patterns and Relations

A pattern starts at 100 and each step subtracts 15. What are the first four numbers of this pattern?

Answer

The first four numbers are 100, 85, 70, 55. Subtract 15 from each term to get the next one: $100 - 15 = 85$, $85 - 15 = 70$, $70 - 15 = 55$.

#199 · Geometry and Measurement

A right triangle has a hypotenuse of 13 cm and one leg of 5 cm. What is the length of the other leg?

Answer

The other leg is 12 cm. Use $a^2 + b^2 = c^2$, so $5^2 + b^2 = 13^2$: $25 + b^2 = 169$, so $b^2 = 144$ and $b = \text{the square root of } 144 = 12 \text{ cm}$.

#200 · Data Management and Probability

The probability that it rains tomorrow is 0.2. What is the probability that it does NOT rain?

Answer

The probability is 0.8. The chance of rain and the chance of no rain must add to 1, so subtract: $1 - 0.2 = 0.8$. This is the complement, meaning the opposite outcome.

#201 · Number Sense and Operations

What is $\frac{3}{4}$ written as a decimal?

Answer

0.75. Divide the numerator by the denominator: 3 divided by 4 is 0.75. The numerator is the top number and the denominator is the bottom number.

#202 · Patterns and Relations

What is the next number in this pattern: 2, 5, 8, 11, ...?

Answer

14. The pattern adds 3 each time: 2 plus 3 is 5, 5 plus 3 is 8, 8 plus 3 is 11, and 11 plus 3 is 14.

#203 · Geometry and Measurement

What is the perimeter of a rectangle that is 8 cm long and 5 cm wide?

Answer

26 cm. Perimeter is the distance around the shape. Add the length and the width, then double the total: 2 times (8 plus 5) is 2 times 13, which is 26.

#204 · Data Management and Probability

What is the mean of 4, 8, 6, and 10?

Answer

7. The mean is the average. Add the numbers to get 28, then divide by how many numbers there are, which is 4: 28 divided by 4 is 7.

#205 · Number Sense and Operations

Write 0.45 as a percent.

Answer

45%. Percent means out of 100. Move the decimal point two places to the right, then add a percent sign: 0.45 becomes 45%.

#206 · Patterns and Relations

Solve for x: x plus 7 equals 15.

Answer

$x = 8$. To get x alone, do the opposite of adding 7, which is subtracting 7, on both sides: 15 minus 7 is 8.

#207 · Geometry and Measurement

What is the area of a rectangle that is 8 cm long and 5 cm wide?

Answer

40 square cm. Area is the space inside the shape. For a rectangle, multiply the length by the width: 8 times 5 is 40.

#208 · Data Management and Probability

What is the median of 3, 7, 2, 9, and 5?

Answer

5. The median is the middle value. Put the numbers in order first: 2, 3, 5, 7, 9. The middle number is 5.

#209 · Number Sense and Operations

What is 20% of 150?

Answer

30. Percent means out of 100, so 20% is $20/100$, which is 0.20. Multiply 0.20 by 150: 0.20 times 150 is 30.

#210 · Patterns and Relations

Solve for x: 3x plus 4 equals 19.

Answer

$x = 5$. Subtract 4 from both sides to get $3x = 15$. Then divide both sides by 3: 15 divided by 3 is 5.

#211 · Geometry and Measurement

What is the area of a triangle with a base of 10 cm and a height of 6 cm?

Answer

30 square cm. For a triangle, multiply the base by the height, then take half: (10 times 6) is 60, and 60 divided by 2 is 30.

#212 · Data Management and Probability

What is the mode of 2, 4, 4, and 6?

Answer

4. The mode is the value that appears most often. In this list, 4 appears twice while 2 and 6 appear once, so the mode is 4.

#213 · Number Sense and Operations

What is $\frac{1}{3}$ plus $\frac{1}{4}$?

Answer

$\frac{7}{12}$. To add fractions with different bottom numbers, first find a common denominator. 3 and 4 both go into 12, so $\frac{1}{3}$ is $\frac{4}{12}$ and $\frac{1}{4}$ is $\frac{3}{12}$. Add the top numbers: 4 plus 3 is 7, so the total is $\frac{7}{12}$.

#214 · Patterns and Relations

Simplify the ratio 12 to 18.

Answer

2 to 3. A ratio compares two amounts. Divide both parts by their greatest common factor, which is 6: 12 divided by 6 is 2, and 18 divided by 6 is 3.

#215 · Geometry and Measurement

What is the circumference of a circle with a radius of 5 cm? Use 3.14 for pi.

Answer

31.4 cm. Circumference is the distance around a circle. The formula is 2 times pi times the radius: 2 times 3.14 times 5 is 31.4.

#216 · Data Management and Probability

What is the range of 10, 3, 8, and 5?

Answer

7. The range is the largest value minus the smallest: 10 minus 3 is 7.

#217 · Number Sense and Operations

What is 9 minus negative 4?

Answer

13. Subtracting a negative number is the same as adding that number, so 9 plus 4 is 13.

#218 · Patterns and Relations

Solve the proportion: $\frac{3}{5}$ equals $\frac{x}{15}$.

Answer

$x = 9$. A proportion says two fractions are equal. Cross multiply: 3 times 15 is 45, and 5 times x is $5x$, so 45 equals $5x$. Divide both sides by 5: 45 divided by 5 is 9.

#219 · Geometry and Measurement

What is the area of a circle with a radius of 5 cm? Use 3.14 for pi.

Answer

78.5 square cm. The formula is pi times the radius squared, which means radius times itself: 3.14 times 5 times 5 is 3.14 times 25, which is 78.5.

#220 · Data Management and Probability

A data set is 2, 4, 6, 8, and 100. How does the value 100 affect the mean?

Answer

It pulls the mean up. The mean of the other four numbers is (2 plus 4 plus 6 plus 8) divided by 4, which is 5. Adding 100 gives a much higher mean of (2 plus 4 plus 6 plus 8 plus 100) divided by 5, which is 24.

#221 · Number Sense and Operations

What is 3 plus 4 times 2?

Answer

11. Use the order of operations, which is also called BEDMAS, and multiply before you add: 4 times 2 is 8, then 3 plus 8 is 11.

#222 · Patterns and Relations

A car travels 240 km in 3 hours. What is its average speed?

Answer

80 km per hour. Average speed is the distance divided by the time: 240 divided by 3 is 80.

#223 · Geometry and Measurement

What is the volume of a rectangular prism that is 4 cm long, 3 cm wide, and 2 cm tall?

Answer

24 cubic cm. Volume is the space a 3D shape takes up. Multiply the length by the width by the height: 4 times 3 times 2 is 24.

#224 · Data Management and Probability

You roll one six-sided die. What is the probability of rolling an even number?

Answer

1 out of 2. Probability is the number of desired outcomes divided by the total outcomes. The even numbers on a die are 2, 4, and 6, which is 3 outcomes out of 6, so the probability is $\frac{3}{6}$, which simplifies to $\frac{1}{2}$.

#225 · Number Sense and Operations

A jacket costs \$60 and is 20% off. What is the sale price?

Answer

\$48. The discount is 20% of \$60, which is 0.20 times 60, or \$12. Subtract the discount from the original price: 60 minus 12 is 48.

#226 · Patterns and Relations

On a map, 1 cm represents 5 km. How far apart are two towns that are 4 cm apart on the map?

Answer

20 km. Multiply the map distance by the scale: 4 times 5 is 20.

#227 · Geometry and Measurement

**A right triangle has legs of 3 cm and 4 cm.
What is the length of the hypotenuse?**

Answer

5 cm. Use the Pythagorean theorem: leg squared plus leg squared equals hypotenuse squared. So 3 squared is 9 and 4 squared is 16, and 9 plus 16 is 25. The square root of 25 is 5.

#228 · Data Management and Probability

**You flip a coin twice. What is the probability
of getting heads both times?**

Answer

1 out of 4. The two flips are independent, meaning the first does not change the second. Multiply the two probabilities: $\frac{1}{2}$ times $\frac{1}{2}$ is $\frac{1}{4}$.

#229 · Number Sense and Operations

A bill is \$200 before tax and the tax rate is 13%. What is the total with tax?

Answer

\$226. The tax is 13% of \$200, which is 0.13 times 200, or \$26. Add the tax to the bill: 200 plus 26 is 226.

#230 · Patterns and Relations

The area of a rectangle is length times width. If the area is 24 and the width is 6, what is the length?

Answer

4. Replace the known values in the formula: 24 equals length times 6. Divide both sides by 6 to solve for the length: 24 divided by 6 is 4.

#231 · Geometry and Measurement

What is the surface area of a cube with a side length of 3 cm?

Answer

54 square cm. Surface area is the total area of all the faces. A cube has 6 identical square faces. Each face has area 3 times 3, which is 9, so the total is 6 times 9, which is 54.

#232 · Data Management and Probability

You roll one six-sided die. What is the probability of rolling a 2 or a 5?

Answer

1 out of 3. Rolling a 2 and rolling a 5 cannot happen at the same time, so the events are mutually exclusive. Add the two probabilities: $\frac{1}{6}$ plus $\frac{1}{6}$ is $\frac{2}{6}$, which simplifies to $\frac{1}{3}$.

#233 · Number Sense and Operations

What is $\frac{3}{4}$ of 200?

Answer

150. A fraction of a number means multiply. First divide 200 by the bottom number 4 to get 50, then multiply by the top number 3: 50 times 3 is 150.

#234 · Patterns and Relations

What is the next number in this pattern: 3, 6, 12, 24, ...?

Answer

48. The pattern multiplies by 2 each time: 3 times 2 is 6, 6 times 2 is 12, 12 times 2 is 24, and 24 times 2 is 48.

#235 · Geometry and Measurement

What is the area of a parallelogram with a base of 12 cm and a height of 5 cm?

Answer

60 square cm. The area of a parallelogram is the base times the height: 12 times 5 is 60.

#236 · Data Management and Probability

In a class of 20 students, 8 like blue, 7 like red, and the rest like green. How many students like green?

Answer

5. Add the students who like blue and red: 8 plus 7 is 15. Then subtract that from the total: 20 minus 15 is 5.

#237 · Number Sense and Operations

**A meal costs \$50 and you leave a 15% tip.
How much is the tip?**

Answer

\$7.50. To find 15% of \$50, multiply 0.15 by 50: 0.15 times 50 is 7.5, so the tip is \$7.50.

#238 · Patterns and Relations

Solve for x: 4x equals 28.

Answer

$x = 7$. Divide both sides by 4 to get x alone: 28 divided by 4 is 7.

#239 · Geometry and Measurement

Convert 2.5 metres to centimetres.

Answer

250 cm. There are 100 centimetres in 1 metre, so multiply the metres by 100: 2.5 times 100 is 250.

#240 · Data Management and Probability

A circle graph shows that one section is 25% of the whole. How many degrees does that section take up?

Answer

90 degrees. A circle has 360 degrees in total. Find 25% of 360: 0.25 times 360 is 90.

#241 · Number Sense and Operations

A salesperson earns 8% commission on \$2,500 in sales. How much commission do they earn?

Answer

\$200. Commission is a percent of the sales amount: 0.08 times 2500 is \$200.

#242 · Patterns and Relations

Solve for x: x minus 6 equals 4.

Answer

$x = 10$. Add 6 to both sides to get x alone: 4 plus 6 is 10.

#243 • Geometry and Measurement

What is the perimeter of a square with a side length of 9 cm?

Answer

36 cm. A square has 4 equal sides, so the perimeter is 4 times the side length: 4 times 9 is 36.

#244 • Data Management and Probability

A scatterplot of ice cream sales against temperature rises from left to right. What does this tell you?

Answer

There is a positive relationship. As the temperature goes up, the ice cream sales also go up, so the two values increase together.

#245 · Number Sense and Operations

You put \$1,000 in a savings account at 5% simple interest per year. How much interest do you earn in 3 years?

Answer

\$150. Simple interest is the principal times the rate times the time. The principal is \$1,000, the rate is 0.05, and the time is 3 years: $1000 \times 0.05 \times 3$ is \$150.

#246 · Patterns and Relations

Using the formula distance equals rate times time, how far do you travel if you go 60 km per hour for 2 hours?

Answer

120 km. Substitute the known values into the formula: rate is 60 and time is 2, so 60×2 is 120.

#247 • Geometry and Measurement

A photo is 12 cm wide and is enlarged by a scale factor of 2.5. How wide is the new photo?

Answer

30 cm. A scale factor tells you how many times bigger the new size is. Multiply the original width by the scale factor: 12 times 2.5 is 30.

#248 • Data Management and Probability

A line graph shows a temperature of 15 degrees at 9 am and 18 degrees at 10 am. By how many degrees did the temperature rise?

Answer

3 degrees. Find the change by subtracting the earlier reading from the later reading: 18 minus 15 is 3.

#249 · Number Sense and Operations

What is $\frac{1}{2}$ divided by $\frac{1}{4}$?

Answer

2. To divide fractions, keep the first fraction, flip the second fraction, and then multiply: $\frac{1}{2}$ times $\frac{4}{1}$ is $\frac{4}{2}$, which equals 2.

#250 · Patterns and Relations

Share \$100 between two people in the ratio 2 to 3. How much does each person get?

Answer

\$40 and \$60. Add the ratio parts: 2 plus 3 is 5. Each part is \$100 divided by 5, or \$20. The first person gets 2 parts, so 2 times 20 is \$40, and the second gets 3 parts, so 3 times 20 is \$60.

#251 · Geometry and Measurement

What is the perimeter of a rectangle that is 8 cm long and 5 cm wide?

Answer

26 cm. Add all four sides: $8 + 5 + 8 + 5 = 26$. Or double the sum of length and width: $2 \times (8 + 5) = 26$.

#252 · Data Management and Probability

What is the mean of the numbers 4, 6, 8, and 10?

Answer

7. Add the numbers to get 28, then divide by how many numbers there are: $28 \div 4 = 7$.

#253 · Number Sense and Operations

Convert the fraction $\frac{3}{8}$ to a decimal.

Answer

0.375. To change a fraction to a decimal, divide the top number by the bottom number: 3 divided by 8 = 0.375.

#254 · Patterns and Relations

The pattern is 5, 8, 11, 14. What is the next term, and what is the rule?

Answer

The next term is 17. The rule is add 3 each time, so $14 + 3 = 17$. Each term grows by the same amount, so it is an arithmetic pattern.

#255 · Geometry and Measurement

What is the area of a rectangle that is 8 cm long and 5 cm wide?

Answer

40 square centimetres. Area of a rectangle equals length times width: $8 \times 5 = 40$.

#256 · Data Management and Probability

What is the median of the numbers 3, 8, 5, 9, and 2?

Answer

5. Put the numbers in order: 2, 3, 5, 8, 9. The median is the middle value, which is 5.

#257 · Number Sense and Operations

Convert the fraction $\frac{3}{5}$ to a percent.

Answer

60%. First divide 3 by 5 to get 0.6, then multiply by 100 to change the decimal to a percent: $0.6 \times 100 = 60\%$.

#258 · Patterns and Relations

The pattern is 2, 4, 6, 8. What rule gives the next term?

Answer

Add 2 to the previous term, so $8 + 2 = 10$. This is the list of even numbers, and each term is 2 more than the one before it.

#259 · Geometry and Measurement

What is the area of a triangle with a base of 10 m and a height of 6 m?

Answer

30 square metres. Area of a triangle equals one half times base times height: $\frac{1}{2} \times 10 \times 6 = 30$.

#260 · Data Management and Probability

What is the mode of the list 2, 5, 5, 7, 5, 9?

Answer

5. The mode is the number that appears most often. In this list 5 appears three times, more than any other number.

#261 · Number Sense and Operations

What is $\frac{1}{4}$ plus $\frac{1}{6}$?

Answer

$\frac{5}{12}$. The common denominator is 12. Convert $\frac{1}{4}$ to $\frac{3}{12}$ and $\frac{1}{6}$ to $\frac{2}{12}$, then add the tops: $\frac{3}{12} + \frac{2}{12} = \frac{5}{12}$.

#262 · Patterns and Relations

Solve for x: $x + 7 = 15$.

Answer

$x = 8$. Subtract 7 from both sides to get x alone: 15 minus $7 = 8$.

#263 · Geometry and Measurement

What is the circumference of a circle with a diameter of 10 cm? Use 3.14 for pi.

Answer

31.4 cm. Circumference equals pi times diameter:
 $3.14 \times 10 = 31.4$.

#264 · Data Management and Probability

What is the range of the data set 4, 9, 2, 12, and 6?

Answer

10. The range is the largest value minus the smallest value: $12 \text{ minus } 2 = 10$.

#265 · Number Sense and Operations

What is 4 plus 3 times 2?

Answer

10. Do the multiplication before the addition: $3 \times 2 = 6$, then $4 + 6 = 10$. This is the order of operations, BEDMAS.

#266 · Patterns and Relations

Solve for x: $x + 3 + 4 = 12$.

Answer

$x = 5$. Add 3 and 4 to get 7, then subtract 7 from 12: 12 minus 7 = 5.

#267 · Geometry and Measurement

What is the area of a circle with a radius of 3 cm? Use 3.14 for pi.

Answer

28.26 square centimetres. Area equals pi times the radius squared: $3.14 \times 3 \times 3 = 3.14 \times 9 = 28.26$.

#268 · Data Management and Probability

A data set has a mean of 50, then a very high value of 200 is added. Does the mean or the median change more?

Answer

The mean changes more. The mean uses every value, so one very high number pulls it up. The median only moves to the middle value, so it is barely affected.

#269 · Number Sense and Operations

What is 15% of 80?

Answer

12. Change 15% to 0.15, then multiply: $0.15 \times 80 =$
12. Percent means out of 100.

#270 · Patterns and Relations

Solve for x: $3x = 21$.

Answer

$x = 7$. Divide both sides by 3 to get x alone: 21
divided by $3 = 7$.

#271 · Geometry and Measurement

What is the volume of a box that is 4 cm long, 3 cm wide, and 2 cm tall?

Answer

24 cubic centimetres. Volume of a rectangular prism equals length times width times height: $4 \times 3 \times 2 = 24$.

#272 · Data Management and Probability

A bag has 3 red balls and 2 blue balls. What is the probability of drawing a red ball?

Answer

$\frac{3}{5}$, or 0.6. There are 5 balls total and 3 are red, so the probability is 3 out of 5, which is 3 divided by 5 = 0.6.

#273 · Number Sense and Operations

A price rises from \$50 to \$60. What is the percent increase?

Answer

20%. The increase is 10. Divide the increase by the original amount: $10 \text{ divided by } 50 = 0.2$, then multiply by 100 to get 20%.

#274 · Patterns and Relations

Solve for x : $2x + 5 = 13$.

Answer

$x = 4$. Subtract 5 from 13 to get 8, then divide 8 by 2: $8 \text{ divided by } 2 = 4$.

#275 · Geometry and Measurement

A right triangle has legs of 3 and 4. How long is the hypotenuse?

Answer

5. Use Pythagoras: $a^2 + b^2 = c^2$. So $9 + 16 = 25$, and the square root of 25 is 5.

#276 · Data Management and Probability

If the probability of rain is 30%, what is the probability that it does not rain?

Answer

70%. Rain and no rain are opposites, so they add to 100%: $100 - 30 = 70$.

#277 · Number Sense and Operations

A \$40 jacket is 25% off. What is the sale price?

Answer

\$30. The discount is 25% of 40, which is $0.25 \times 40 = 10$. Subtract the discount from the original price: $40 \text{ minus } 10 = 30$.

#278 · Patterns and Relations

**Speed equals distance divided by time.
How fast is a car that travels 240 km in 3 hours?**

Answer

80 km/h. Divide the distance by the time: $240 \text{ divided by } 3 = 80$.

#279 · Geometry and Measurement

A right triangle has a hypotenuse of 13 and one leg of 5. How long is the other leg?

Answer

12. Use Pythagoras: $c^2 = a^2 + b^2$. So $13^2 = 5^2 + b^2$, which gives $169 = 25 + b^2$, and $b^2 = 144$, and the square root of 144 is 12.

#280 · Data Management and Probability

You flip a coin and roll a die. What is the probability of getting heads and a 6?

Answer

$\frac{1}{12}$, which is about 0.08. The two events are independent, so multiply: $\frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$.

#281 · Number Sense and Operations

An item costs \$20 before tax. With 13% sales tax, what is the total cost?

Answer

\$22.60. The tax is 13% of 20, which is $0.13 \times 20 = 2.60$. Add the tax to the price: $20 + 2.60 = 22.60$.

#282 · Patterns and Relations

Use the formula $A = l \times w$. What is the area of a rectangle with a length of 9 m and a width of 4 m?

Answer

36 square metres. Put the numbers into the formula: $A = 9 \times 4 = 36$.

#283 · Geometry and Measurement

A model car is $\frac{1}{20}$ the size of the real car. If the real car is 400 cm long, how long is the model?

Answer

20 cm. To scale down, divide the real length by the scale factor: $400 \text{ divided by } 20 = 20$.

#284 · Data Management and Probability

A bag has 4 red, 3 blue, and 3 green marbles. What is the probability of drawing a red or a blue marble?

Answer

$\frac{7}{10}$, or 0.7. Red and blue cannot happen at the same time, so add their counts: $4 + 3 = 7$, out of 10 marbles total.

#285 · Number Sense and Operations

You invest \$500 at 4% simple interest for 1 year. How much interest do you earn?

Answer

\$20. Simple interest equals principal times rate times time: $500 \times 0.04 \times 1 = 20$.

#286 · Patterns and Relations

If 4 apples cost \$2, how much do 10 apples cost?

Answer

\$5. Find the unit rate first: 2 divided by 4 = 0.50 per apple. Then multiply by 10: $0.50 \times 10 = 5$.

#287 · Geometry and Measurement

How many millilitres are in 2.5 litres?

Answer

2,500 mL. Each litre holds 1,000 mL, so multiply 2.5 by 1000 = 2,500.

#288 · Data Management and Probability

A circle graph shows that 25% of students chose music. If there are 200 students, how many chose music?

Answer

50 students. Take 25% of the total: $0.25 \times 200 = 50$.

#289 · Number Sense and Operations

**A meal costs \$45 and you leave a 15% tip.
How much is the tip?**

Answer

\$6.75. Change 15% to 0.15 and multiply by the bill:
 $0.15 \times 45 = 6.75$.

#290 · Patterns and Relations

**The distance formula is $d = r \times t$. Find t
when $d = 60$ and $r = 12$.**

Answer

$t = 5$. Rearrange the formula to $t = d$ divided by r ,
then divide: 60 divided by $12 = 5$.

#291 • Geometry and Measurement

What is the volume of a cylinder with a radius of 2 cm and a height of 5 cm? Use 3.14 for pi.

Answer

62.8 cubic centimetres. Volume equals pi times radius squared times height: $3.14 \times 2 \times 2 \times 5 = 3.14 \times 4 \times 5 = 62.8$.

#292 • Data Management and Probability

A bar graph shows Monday sold 10, Tuesday 20, and Wednesday 15. How many were sold in total?

Answer

45. Add the three values: $10 + 20 + 15 = 45$.

#293 · Number Sense and Operations

A salesperson earns 5% commission on a \$2,000 sale. How much commission do they earn?

Answer

\$100. Change 5% to 0.05 and multiply by the sale amount: $0.05 \times 2000 = 100$.

#294 · Patterns and Relations

A number plus 6 equals 14. What is the number?

Answer

8. Write the equation $x + 6 = 14$, then subtract 6 from both sides: $14 \text{ minus } 6 = 8$.

#295 • Geometry and Measurement

What is the surface area of a box that is 4 cm long, 3 cm wide, and 2 cm tall?

Answer

52 square centimetres. Add the areas of all six faces: $2 \times (4 \times 3) + 2 \times (4 \times 2) + 2 \times (3 \times 2) = 24 + 16 + 12 = 52$.

#296 • Data Management and Probability

A table shows town A had 120 people and town B had 80. How many more people does town A have?

Answer

40 more people. Subtract the smaller from the larger: $120 \text{ minus } 80 = 40$.

#297 · Number Sense and Operations

Convert the mixed number $2\frac{3}{4}$ to an improper fraction.

Answer

$\frac{11}{4}$. Multiply the whole number by the denominator ($2 \times 4 = 8$), add the numerator ($8 + 3 = 11$), and keep the denominator 4.

#298 · Patterns and Relations

In a table, each input is doubled to make the output: 1 goes to 2, 2 goes to 4, 3 goes to 6. What output goes with input 5?

Answer

10. Multiply the input by 2 to get the output: $5 \times 2 = 10$.

#299 · Geometry and Measurement

A rectangle is 8 cm by 4 cm with a 2 cm by 2 cm square cut out. What is the area of the remaining shape?

Answer

28 square centimetres. The rectangle area is $8 \times 4 = 32$, and the square is $2 \times 2 = 4$. Subtract: 32 minus $4 = 28$.

#300 · Data Management and Probability

A scatterplot shows that as years of study increase, test scores also rise. What kind of trend does this show?

Answer

A positive trend. When both values go up together, the points slope upward, which shows a positive relationship between the two.

#301 · Number Sense and Operations

Write 40% as a fraction in lowest terms.

Answer

The fraction is $\frac{2}{5}$. A percent means out of 100, so write 40 over 100. Divide the top and bottom by the greatest common factor, which is 20, to get $\frac{40}{100} = \frac{2}{5}$.

#302 · Geometry and Measurement

A square has a side length of 6 cm. What is its perimeter?

Answer

The perimeter is 24 cm. A square has four equal sides, so multiply the side by 4: $6 \times 4 = 24$ cm. Perimeter is the distance all the way around a shape.

#303 · Data Management and Probability

A student scored 70, 80, 25, and 85 on four tests. What is the mean score?

Answer

The mean is 65. Add the scores ($70 + 80 + 25 + 85 = 260$), then divide by how many tests there are, which is 4: $260 \div 4 = 65$. The low score of 25 pulls the mean down.

#304 · Number Sense and Operations

Evaluate $5 + 2$ to the power of 2 times 3. Show the order of operations.

Answer

The answer is 17. First do the power: $2^2 = 4$. Then do the multiplication: $4 \times 3 = 12$. Finally do the addition: $5 + 12 = 17$.

#305 · Geometry and Measurement

A parallelogram has a base of 9 cm and a height of 4 cm. What is its area?

Answer

The area is 36 square cm. Use the formula $\text{area} = \text{base} \times \text{height}$: $9 \times 4 = 36$. The height must be the perpendicular distance from the base to the top side.

#306 · Data Management and Probability

The numbers are 12, 7, 15, and 9. What is the median?

Answer

The median is 10.5. Put the numbers in order: 7, 9, 12, 15. With an even count, average the two middle numbers: $(9 + 12) \div 2 = 10.5$.

#307 · Number Sense and Operations

A jacket costs \$40 and is 30% off. What is the sale price?

Answer

The sale price is \$28. The discount is 30% of \$40, or $0.30 \times 40 = \$12$. Subtract the discount from the original price: $\$40 - \$12 = \$28$.

#308 · Geometry and Measurement

A box is 6 cm long, 4 cm wide, and 3 cm tall. What is its volume?

Answer

The volume is 72 cubic cm. Multiply length by width by height: $6 \times 4 \times 3 = 72$. Volume measures how much space the box takes up.

#309 · Data Management and Probability

A circle graph has four sections of 30%, 40%, 20%, and one unknown section. What percent is the unknown section?

Answer

The unknown section is 10%. A circle graph always adds to 100%. Add the known sections ($30 + 40 + 20 = 90$), then subtract from 100: $100 - 90 = 10$.

#310 · Number Sense and Operations

Write the improper fraction $13/4$ as a mixed number.

Answer

The mixed number is $3 \frac{1}{4}$. Divide the numerator (top) by the denominator (bottom): 13 divided by 4 = 3 with a remainder of 1. The 3 is the whole number, the 1 becomes the new numerator, and the denominator stays 4.

#311 · Geometry and Measurement

What is the area of a circle with a radius of 6 cm? Use $\pi = 3.14$.

Answer

The area is 113.04 square cm. Use $\text{area} = \pi \times \text{radius squared}$: $3.14 \times 6 \times 6 = 3.14 \times 36 = 113.04$. The radius is the distance from the centre to the edge of the circle.

#312 · Data Management and Probability

In a class of 25 students, 10 play soccer. If one student is chosen at random, what is the probability they play soccer?

Answer

The probability is $\frac{2}{5}$. Probability = favourable outcomes divided by total outcomes, so 10 out of 25. Reduce the fraction by dividing top and bottom by 5: $\frac{10}{25} = \frac{2}{5}$.

#313 · Number Sense and Operations

A meal costs \$50 before tax, and sales tax is 15%. What is the total cost?

Answer

The total is \$57.50. The tax is 15% of \$50, or $0.15 \times 50 = \$7.50$. Add the tax to the original price: $\$50 + \$7.50 = \$57.50$.

#314 · Geometry and Measurement

A right triangle has legs of 6 cm and 8 cm. How long is the hypotenuse?

Answer

The hypotenuse is 10 cm. Use the Pythagorean theorem: leg squared + leg squared = hypotenuse squared. 6 squared = 36 and 8 squared = 64, so $36 + 64 = 100$. The square root of 100 is 10.

#315 · Data Management and Probability

You flip a coin and roll a six-sided die. What is the probability of getting tails and a 3?

Answer

The probability is $\frac{1}{12}$. The two events are independent, so multiply their separate chances: $\frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$. Independent events do not affect each other.

#316 · Number Sense and Operations

Put the integers -5, 2, -1, and 0 in order from smallest to largest.

Answer

The order is -5, -1, 0, 2. On a number line, a negative number to the left is smaller, so -5 is the smallest and -1 comes before 0. Zero is bigger than any negative number but smaller than 2.

#317 • Geometry and Measurement

A circle has a diameter of 12 cm. What is its circumference? Use $\pi = 3.14$.

Answer

The circumference is 37.68 cm. Use circumference = $\pi \times \text{diameter}$: $3.14 \times 12 = 37.68$. The diameter is the distance across the circle through the centre.

#318 • Data Management and Probability

The chance of drawing a heart is $\frac{1}{4}$ and the chance of drawing a spade is $\frac{1}{4}$. What is the probability of drawing a heart or a spade?

Answer

The probability is $\frac{1}{2}$. A card cannot be both a heart and a spade, so the events are mutually exclusive. Add the chances: $\frac{1}{4} + \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$.

#319 • Number Sense and Operations

**A meal costs \$45 and you leave a 20% tip.
How much is the tip?**

Answer

The tip is \$9. Convert 20% to a decimal (0.20), then multiply by the bill: $0.20 \times 45 = 9$. A tip is a percent of the bill added to the cost of the meal.

#320 • Geometry and Measurement

**A box is 5 cm long, 4 cm wide, and 3 cm tall.
What is its surface area?**

Answer

The surface area is 94 square cm. The three different faces have areas $5 \times 4 = 20$, $5 \times 3 = 15$, and $4 \times 3 = 12$. Add them ($20 + 15 + 12 = 47$) and double for the opposite faces: $2 \times 47 = 94$.

#321 · Data Management and Probability

The numbers are 4, 7, 4, 9, 7, and 4. What is the mode?

Answer

The mode is 4. The mode is the value that appears most often, and 4 appears three times while 7 appears twice and 9 appears once.

#322 · Number Sense and Operations

What is $\frac{3}{4}$ of 160?

Answer

The answer is 120. For a fraction of a number, divide by the denominator (bottom) then multiply by the numerator (top): $160 \div 4 = 40$, then $40 \times 3 = 120$.

#323 · Geometry and Measurement

A shape is enlarged by a scale factor of 4. A side that was 3 cm long becomes how long?

Answer

The new side is 12 cm. Multiply the original measurement by the scale factor: $3 \times 4 = 12$. A scale factor greater than 1 makes the shape larger, and every side is multiplied by the same factor.

#324 · Data Management and Probability

The daily high temperatures were 22, 18, 25, and 20 degrees. What is the range?

Answer

The range is 7 degrees. Subtract the smallest value from the largest: $25 - 18 = 7$. The range shows how spread out the temperatures were.

#325 · Number Sense and Operations

Write $\frac{3}{5}$ as a percent.

Answer

The percent is 60%. Divide the numerator (top) by the denominator (bottom) to get a decimal: $3 \div 5 = 0.60$. Then multiply by 100 to get a percent: $0.60 \times 100 = 60\%$.

#326 · Geometry and Measurement

A cylinder has a radius of 3 cm and a height of 5 cm. What is its volume? Use $\pi = 3.14$.

Answer

The volume is 141.3 cubic cm. Use volume = $\pi \times \text{radius squared} \times \text{height}$: $3.14 \times 3 \times 3 \times 5 = 3.14 \times 45 = 141.3$. The volume is the area of the circular end times the height.

#327 · Data Management and Probability

A number puzzle says: pick a number, multiply it by 3, then subtract 4. If the result is 14, what was the number?

Answer

The number is 6. Set up the equation $3x - 4 = 14$. Add 4 to both sides to get $3x = 18$, then divide both sides by 3 to get $x = 6$. Use a letter like x for the unknown number.

#328 · Number Sense and Operations

Round the decimal 4.736 to the nearest hundredth.

Answer

The rounded number is 4.74. The hundredths place is the second digit after the decimal point, which is 3. Look at the next digit, 6, and round 3 up to 4 because 6 is 5 or more.

#329 · Geometry and Measurement

Convert 3.2 metres to centimetres.

Answer

The answer is 320 cm. One metre equals 100 centimetres, so multiply 3.2 by 100: $3.2 \times 100 = 320$. Moving from a larger unit to a smaller one means multiplying.

#330 · Data Management and Probability

A bar graph shows books read: Alex 5, Maria 8, Sam 3. What is the total, and who read the most?

Answer

The total is 16 books, and Maria read the most. Add the three bar values: $5 + 8 + 3 = 16$. The tallest bar belongs to Maria, whose value of 8 is the largest.

#331 · Number Sense and Operations

What is the simple interest on \$800 at 6% per year for 2 years?

Answer

The interest is \$96. Use interest = principal x rate x time: $800 \times 0.06 \times 2 = 96$. The principal is the starting amount, the rate is written as a decimal, and the time is in years.

#332 · Geometry and Measurement

Convert 2.5 kilograms to grams.

Answer

The answer is 2500 grams. One kilogram equals 1000 grams, so multiply 2.5 by 1000: $2.5 \times 1000 = 2500$. A kilogram is a larger unit than a gram.

#333 · Data Management and Probability

A histogram shows that 5 students scored between 60 and 69, and 8 students scored between 70 and 79. How many students are in these two bins together?

Answer

There are 13 students. Add the counts from the two bins: $5 + 8 = 13$. A histogram groups data into ranges, and each bar shows how many items fall in that range.

#334 · Number Sense and Operations

Evaluate $10 - 2 \times 3$ using the order of operations.

Answer

The answer is 4. Do the multiplication before the subtraction: $2 \times 3 = 6$. Then subtract: $10 - 6 = 4$. Working left to right only would give the wrong answer of 24.

#335 · Geometry and Measurement

An L-shaped figure is made of a 5 cm by 4 cm rectangle and a 3 cm by 2 cm rectangle. What is the total area?

Answer

The total area is 26 square cm. Find the area of each rectangle and add them: $5 \times 4 = 20$, and $3 \times 2 = 6$. The total is $20 + 6 = 26$.

#336 · Data Management and Probability

A spinner has red, blue, and green sections. The chance of red is 0.4 and the chance of blue is 0.35. What is the chance of green?

Answer

The chance of green is 0.25. The probabilities of all sections add to 1. Add red and blue ($0.4 + 0.35 = 0.75$), then subtract from 1: $1 - 0.75 = 0.25$.

#337 · Number Sense and Operations

A worker earns \$16 per hour and works 8 hours a day for 5 days. What is the total pay?

Answer

The total pay is \$640. Find the total hours: $8 \times 5 = 40$ hours. Then multiply the hours by the hourly wage: $40 \times 16 = 640$. Pay is the wage per hour times the hours worked.

#338 · Geometry and Measurement

A cube has a side length of 4 cm. What is its surface area?

Answer

The surface area is 96 square cm. One face has area $4 \times 4 = 16$. A cube has six equal faces, so multiply by 6: $6 \times 16 = 96$.

#339 · Data Management and Probability

A line graph shows a stock price of \$15 on Monday and \$21 on Friday. By how much did the price increase?

Answer

The price increased by \$6. Subtract the earlier value from the later value: $21 - 15 = 6$. A rising line on a line graph shows a value going up over time.

#340 · Number Sense and Operations

What is $1\frac{1}{2} + 2\frac{1}{4}$? Give the answer as a mixed number.

Answer

The sum is $3\frac{3}{4}$. Change each mixed number to a fraction: $1\frac{1}{2} = \frac{3}{2} = \frac{6}{4}$, and $2\frac{1}{4} = \frac{9}{4}$. Add the numerators: $6 + 9 = 15$, so $\frac{15}{4} = 3\frac{3}{4}$.

#341 • Geometry and Measurement

A cube has an edge length of 5 cm. What is its volume?

Answer

The volume is 125 cubic cm. Multiply the edge by itself three times: $5 \times 5 \times 5 = 125$. Volume is the space inside a 3D shape.

#342 • Data Management and Probability

A scatterplot shows hours a phone is used against the amount of battery left. The points fall from left to right. What does the pattern show?

Answer

The pattern shows that more hours of use are paired with less battery left. When points fall from left to right, the two values move in opposite directions. This is a negative relationship.

#343 · Number Sense and Operations

A salesperson earns a 6% commission on a \$1,500 sale. What is the commission?

Answer

The commission is \$90. Multiply the sale amount by the commission percent written as a decimal: $0.06 \times 1500 = 90$. Commission is a percent of the sale paid to the salesperson.

#344 · Geometry and Measurement

A recipe needs 2 pounds of flour. Given that 1 pound = 16 ounces, how many ounces is that?

Answer

The answer is 32 ounces. Multiply the number of pounds by the conversion rate: $2 \times 16 = 32$. Because the conversion is provided, multiply the starting amount by the rate.

#345 · Data Management and Probability

A plumber logged these calls per day: 1, 0, 5, and 2. What is the mean number of calls per day?

Answer

The mean is 2 calls per day. Add the four values ($1 + 0 + 5 + 2 = 8$), then divide by 4 days: $8 \div 4 = 2$. A value of 0 still counts as one of the days.

#346 · Number Sense and Operations

A 4-pack of juice costs \$6. What is the price per bottle?

Answer

The price per bottle is \$1.50. Divide the total cost by the number of bottles: $6 \div 4 = 1.50$. A unit price is the cost for one single item.

#347 · Geometry and Measurement

A right triangle has a hypotenuse of 10 cm and one leg of 6 cm. How long is the other leg?

Answer

The other leg is 8 cm. Use the Pythagorean theorem: leg squared + leg squared = hypotenuse squared, so the missing leg squared = $100 - 36 = 64$. The square root of 64 is 8.

#348 · Data Management and Probability

The numbers are 17, 12, 25, 19, and 10. What is the median?

Answer

The median is 17. Put the numbers in order first: 10, 12, 17, 19, 25. With an odd count, the median is the middle number, which is 17.

#349 · Number Sense and Operations

What is 6 minus negative 3?

Answer

The answer is 9. Subtracting a negative is the same as adding the positive value: $6 - (-3) = 6 + 3 = 9$. The two minus signs cancel out to become a plus.

#350 · Geometry and Measurement

A map scale says 1 cm = 4 km. Two towns are 7 cm apart on the map. How far apart are they in real life?

Answer

They are 28 km apart. Multiply the distance on the map by the scale factor: $7 \times 4 = 28$. The scale factor tells you how many kilometres each centimetre represents.

#351 · Data Management and Probability

What is the mean of the numbers 5, 10, 15, and 20?

Answer

The mean is 12.5. The mean is the average. Add all the numbers ($5 + 10 + 15 + 20 = 50$) and divide by how many there are, which is 4: 50 divided by $4 = 12.5$.

#352 · Number Sense and Operations

How do you write the decimal 0.35 as a fraction in simplest form?

Answer

The fraction is $\frac{7}{20}$. The decimal 0.35 is 35 hundredths, so write it as $\frac{35}{100}$. Divide the top and bottom by 5, and 35 divided by 5 is 7 while 100 divided by 5 is 20.

#353 · Geometry and Measurement

**A rectangle is 12 m long and 5 m wide.
What is its perimeter?**

Answer

The perimeter is 34 m. Perimeter is the distance around a shape. Add the length and width ($12 + 5 = 17$) and double the total to cover two of each side: $2 \times 17 = 34$.

#354 · Data Management and Probability

The numbers are 3, 5, 7, and 9. What is the median?

Answer

The median is 6. The list is already in order and has an even count, so find the two middle numbers, 5 and 7, and average them: $(5 + 7) \text{ divided by } 2 = 6$.

#355 · Number Sense and Operations

What is $\frac{1}{4}$ plus $\frac{1}{6}$? Show the common denominator.

Answer

The sum is $\frac{5}{12}$. Use a common denominator of 12, then rewrite each fraction: $\frac{1}{4} = \frac{3}{12}$ and $\frac{1}{6} = \frac{2}{12}$. Add the top numbers: $3 + 2 = 5$, so the answer is $\frac{5}{12}$.

#356 · Geometry and Measurement

A square has a side length of 6 cm. What is its area?

Answer

The area is 36 square cm. Area is the space inside a flat shape. For a square, multiply the side by itself: $6 \times 6 = 36$.

#357 · Data Management and Probability

What is the mode of the data set 2, 4, 4, 4, and 7?

Answer

The mode is 4. The mode is the value that appears most often in a list. The number 4 appears three times, which is more than any other value.

#358 · Number Sense and Operations

How do you write the improper fraction $\frac{19}{6}$ as a mixed number?

Answer

The mixed number is 3 and $\frac{1}{6}$. Divide the top by the bottom: 19 divided by 6 gives 3 with a remainder of 1. The 3 is the whole part, and the remainder 1 sits over the 6.

#359 · Geometry and Measurement

A triangle has a base of 14 cm and a height of 6 cm. What is its area?

Answer

The area is 42 square cm. The area of a triangle is one half times the base times the height: $0.5 \times 14 \times 6 = 42$. The base and the height should meet at a right angle.

#360 · Data Management and Probability

What is the range of the data set 12, 5, 20, and 9?

Answer

The range is 15. The range is the largest value minus the smallest value. The largest is 20 and the smallest is 5, so $20 \text{ minus } 5 = 15$.

#361 · Number Sense and Operations

What is $\frac{2}{3}$ times $\frac{3}{4}$? Simplify the answer.

Answer

The product is $\frac{1}{2}$. Multiply the top numbers ($2 \times 3 = 6$) and the bottom numbers ($3 \times 4 = 12$) to get $\frac{6}{12}$. Divide the top and bottom by 6 to simplify to $\frac{1}{2}$.

#362 · Geometry and Measurement

A circle has a radius of 10 cm. Using pi as 3.14, what is its area?

Answer

The area is 314 square cm. Use area = pi x radius squared: $3.14 \times 10 \times 10$. Since $10 \times 10 = 100$, the working is $3.14 \times 100 = 314$.

#363 · Data Management and Probability

A set of five test scores contains one very low outlier. Which measure of centre is affected more, the mean or the median?

Answer

The mean is affected more. The mean is a total divided by the count, so a very low number drags the total down a lot. The median only finds the middle value, so it stays nearly the same. This is why the median is more reliable when one score is extreme.

#364 · Number Sense and Operations

What is $\frac{2}{5}$ divided by $\frac{4}{5}$?

Answer

The answer is $\frac{1}{2}$. To divide by a fraction, flip the second one and multiply: $\frac{2}{5} \times \frac{5}{4}$. Multiply the tops ($2 \times 5 = 10$) and the bottoms ($5 \times 4 = 20$) to get $\frac{10}{20}$, which simplifies to $\frac{1}{2}$.

#365 • Geometry and Measurement

A circle has a radius of 5 cm. Using pi as 3.14, what is its circumference?

Answer

The circumference is 31.4 cm. Use circumference = $2 \times \pi \times \text{radius}$: $2 \times 3.14 \times 5$. Multiply $2 \times 3.14 = 6.28$, then $6.28 \times 5 = 31.4$.

#366 • Data Management and Probability

On a bar graph the bar labeled Shoes reaches 18 on the vertical count axis. How many shoes were counted?

Answer

18 shoes. The height of a bar shows the value for that category. Since the Shoes bar reaches 18 on the count scale, there are 18 shoes.

#367 · Number Sense and Operations

Evaluate $3 + 6 \times (5 - 2)$ using the order of operations.

Answer

The value is 21. Work inside the brackets first: $5 - 2 = 3$. Then do the multiplication: $6 \times 3 = 18$. Finally add: $3 + 18 = 21$.

#368 · Geometry and Measurement

A right triangle has legs of 6 cm and 8 cm. How long is the hypotenuse?

Answer

The hypotenuse is 10 cm. Use the Pythagorean theorem: leg squared plus leg squared equals hypotenuse squared. $6^2 = 36$ and $8^2 = 64$, and $36 + 64 = 100$. The square root of 100 is 10.

#369 · Data Management and Probability

What does a histogram show, and how is it different from a bar graph?

Answer

A histogram shows how often data values fall into equal ranges, called bins. The bars touch each other because the data is continuous. A bar graph has separate bars with gaps because each bar is a different category.

#370 · Number Sense and Operations

How do you write 20% as a fraction in simplest form?

Answer

The fraction is $\frac{1}{5}$. Percent means out of 100, so 20% is $\frac{20}{100}$. Divide the top and bottom by 20, and 20 divided by 20 is 1 while 100 divided by 20 is 5.

#371 • Geometry and Measurement

A cube has an edge length of 4 cm. What is its surface area?

Answer

The surface area is 96 square cm. A cube has 6 identical square faces. Each face is $4 \times 4 = 16$ square cm, so multiply by 6: $6 \times 16 = 96$.

#372 • Data Management and Probability

A line graph shows March sales of 40 and April sales of 65. By how much did sales increase from March to April?

Answer

Sales increased by 25. Read the value at each month from the vertical axis, then subtract the smaller from the larger: $65 \text{ minus } 40 = 25$.

#373 · Number Sense and Operations

What is 30% of 200?

Answer

The answer is 60. Write 30% as the decimal 0.30, then multiply by 200: $0.30 \times 200 = 60$. You can also think of 30% as 30 per 100, so $30 \times 2 = 60$.

#374 · Geometry and Measurement

**A box is 6 cm long, 5 cm wide, and 3 cm tall.
What is its surface area?**

Answer

The surface area is 126 square cm. The three different face areas are $6 \times 5 = 30$, $6 \times 3 = 18$, and $5 \times 3 = 15$. Add them ($30 + 18 + 15 = 63$) and double for both sides: $2 \times 63 = 126$.

#375 · Data Management and Probability

On a circle graph one sector is labeled 40%. What fraction of the whole does that sector represent?

Answer

The sector represents $\frac{2}{5}$ of the whole. Percent means out of 100, so 40% is $\frac{40}{100}$. Divide the top and bottom by 20 to get $\frac{2}{5}$.

#376 · Number Sense and Operations

A jacket costs \$80 and is on sale for 25% off. What is the sale price?

Answer

The sale price is \$60. The discount is 25% of \$80, or $0.25 \times 80 = \$20$. Take the discount off the original price: $\$80 - \$20 = \$60$.

#377 · Geometry and Measurement

A rectangular prism is 8 cm long, 4 cm wide, and 2 cm tall. What is its volume?

Answer

The volume is 64 cubic cm. Volume is the space an object takes up. Multiply the length by the width by the height: $8 \times 4 \times 2 = 64$.

#378 · Data Management and Probability

A scatterplot of ice cream sales versus temperature shows points rising from lower left to upper right. What sort of relationship is this?

Answer

This is a positive relationship. As the temperature goes up, the ice cream sales also go up. When two values rise together, the relationship is called positive.

#379 · Number Sense and Operations

A meal costs \$40 before tax. Sales tax is 13%. What is the total cost?

Answer

The total is \$45.20. The tax is 13% of \$40, or $0.13 \times 40 = \$5.20$. Add the tax to the original price: $\$40 + \$5.20 = \$45.20$.

#380 · Geometry and Measurement

A cube has an edge length of 5 cm. What is its volume?

Answer

The volume is 125 cubic cm. For a cube, multiply the edge by itself three times: $5 \times 5 = 25$, then $25 \times 5 = 125$.

#381 · Data Management and Probability

A fair spinner has 4 equal sections: red, blue, green, and yellow. What is the probability of landing on green?

Answer

The probability is $\frac{1}{4}$. There is one green section out of four equal sections. Probability equals the number of favourable outcomes divided by the total possible outcomes.

#382 · Number Sense and Operations

A restaurant bill is \$60. What is a 15% tip?

Answer

The tip is \$9. Write 15% as the decimal 0.15, then multiply by the bill: $0.15 \times 60 = 9$. A tip is a percent of the bill paid to the server.

#383 • Geometry and Measurement

A cylinder has a radius of 3 cm and a height of 5 cm. Using pi as 3.14, what is its volume?

Answer

The volume is 141.3 cubic cm. Use volume = pi x radius squared x height: $3.14 \times 9 \times 5$. First $3.14 \times 9 = 28.26$, then $28.26 \times 5 = 141.3$.

#384 • Data Management and Probability

The probability that a customer makes a purchase is 0.3. What is the probability that the customer does not make a purchase?

Answer

The probability is 0.7. The chance of an event happening and the chance of it not happening must add to 1. Subtract the given probability from 1: $1 \text{ minus } 0.3 = 0.7$.

#385 · Number Sense and Operations

A salesperson earns an 8% commission on a \$1200 sale. How much commission is earned?

Answer

The commission is \$96. Convert 8% to the decimal 0.08, then multiply by the sale amount: $0.08 \times 1200 = 96$. Commission is a percent of the sale paid to the salesperson.

#386 · Geometry and Measurement

A photo is enlarged by a scale factor of 2.5. If an original side is 6 cm, what is the new length?

Answer

The new length is 15 cm. Multiply the original measurement by the scale factor: $6 \times 2.5 = 15$. A scale factor greater than 1 enlarges the shape, and every side is multiplied by the same factor.

#387 · Data Management and Probability

A coin is tossed and a fair 4-section spinner is spun. What is the probability of getting a head and landing on the red section?

Answer

The probability is $\frac{1}{8}$. These are independent events, so multiply their separate chances. The chance of a head is $\frac{1}{2}$ and the chance of red is $\frac{1}{4}$. Multiply: $\frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$.

#388 · Number Sense and Operations

What simple interest is earned on \$400 at 5% per year for 3 years?

Answer

The interest is \$60. Use interest = principal $\times$ rate $\times$ time: $400 \times 0.05 \times 3$. Multiply $400 \times 0.05 = 20$, then $20 \times 3 = 60$. The principal is the starting amount.

#389 · Geometry and Measurement

How many millilitres are in 2.5 litres?

Answer

The answer is 2500 mL. One litre equals 1000 millilitres, so multiply the litres by 1000: $2.5 \times 1000 = 2500$. Moving from a larger unit to a smaller one means multiplying.

#390 · Data Management and Probability

A bag has 5 red, 3 blue, and 2 green marbles. What is the probability of drawing a red or a green marble?

Answer

The probability is $\frac{7}{10}$. These two results cannot happen at the same time, so add the counts and divide by the total. There are $5 + 2 = 7$ red or green marbles out of 10 total, so $\frac{7}{10}$.

#391 · Number Sense and Operations

What is 3 minus 8? Explain using a number line.

Answer

The answer is -5. Start at 3 and move 8 spaces to the left, which crosses zero by 5 and lands at negative 5. Subtracting a bigger number from a smaller one gives a negative result.

#392 · Geometry and Measurement

How many centimetres are in 6.5 metres?

Answer

The answer is 650 cm. One metre equals 100 centimetres, so multiply the metres by 100: $6.5 \times 100 = 650$. Each metre is made of 100 centimetres.

#393 · Data Management and Probability

A table shows books read per day: Monday 4, Tuesday 6, Wednesday 5. What is the total books read and the mean per day?

Answer

The total is 15, and the mean is 5 per day. Add the three daily counts: $4 + 6 + 5 = 15$. Divide the total by the number of days, 3, to get the mean: 15 divided by 3 = 5.

#394 · Number Sense and Operations

Evaluate the integer expression $-2 + 4 \times 3$ using order of operations.

Answer

The value is 10. Do the multiplication first: $4 \times 3 = 12$. Then add it to the negative number: $-2 + 12 = 10$. Following the order of operations is what gives the correct result.

#395 · Geometry and Measurement

A shape is a 6 cm by 4 cm rectangle with a 2 cm by 2 cm square attached. What is the total area?

Answer

The total area is 28 square cm. Find each part separately, then add them. The rectangle is $6 \times 4 = 24$, and the square is $2 \times 2 = 4$. Add them: $24 + 4 = 28$.

#396 · Data Management and Probability

A bag has 4 red and 6 blue marbles. What is the probability of drawing a blue marble?

Answer

The probability is $\frac{3}{5}$. There are 6 blue marbles out of 10 total ($4 + 6 = 10$). Write $\frac{6}{10}$, then divide the top and bottom by 2 to get $\frac{3}{5}$.

#397 · Number Sense and Operations

Solve for x: $\frac{3}{4} = \frac{x}{12}$.

Answer

x equals 9. Cross multiply the two fractions: $3 \times 12 = 4 \times x$, so $36 = 4x$. Divide both sides by 4 to get $x = 9$.

#398 · Geometry and Measurement

A box has a volume of 60 cubic cm, a length of 5 cm, and a width of 3 cm. What is its height?

Answer

The height is 4 cm. Volume equals length x width x height, so $60 = 5 \times 3 \times \text{height}$, or $60 = 15 \times \text{height}$. Divide both sides by 15: 60 divided by 15 = 4.

#399 · Data Management and Probability

A number is doubled and then 4 is added, giving 18. What is the number?

Answer

The number is 7. Let the unknown number be x , so the puzzle says $2x + 4 = 18$. Subtract 4 from both sides to get $2x = 14$, then divide by 2: $x = 7$.

#400 · Number Sense and Operations

An employee earns \$18 per hour and works 35 hours in one week. What is the weekly pay?

Answer

The weekly pay is \$630. Multiply the hourly rate by the hours worked: $\$18 \times 35 = \630 . Wages are the rate per hour times the number of hours worked.

#401 · Geometry and Measurement

**A rectangle is 6 cm long and 4 cm wide.
What is its perimeter?**

Answer

The perimeter is 20 cm. Add the length and the width and double the total: $2 \times (6 + 4) = 2 \times 10 = 20$ cm. Perimeter is the total distance around the shape.

#402 · Data Management and Probability

**The numbers are 12, 15, 18, 21, and 24.
What is the mean?**

Answer

The mean, or average, is 18. Add all the numbers: $12 + 15 + 18 + 21 + 24 = 90$, then divide by how many numbers there are: $90 \text{ divided by } 5 = 18$.

#403 · Number Sense and Operations

Add $\frac{1}{4}$ and $\frac{1}{6}$. Show the steps.

Answer

The sum is $\frac{5}{12}$. Use a common denominator of 12:
 $\frac{1}{4} = \frac{3}{12}$ and $\frac{1}{6} = \frac{2}{12}$. Add the numerators: $3 + 2 = 5$, so the answer is $\frac{5}{12}$.

#404 · Geometry and Measurement

A rectangle is 7 m long and 5 m wide. What is its area?

Answer

The area is 35 square m. Multiply the length by the width: $7 \times 5 = 35$. Area is the space inside the shape, so the units are squared (m squared).

#405 · Data Management and Probability

The numbers are 2, 4, 6, and 8. What is the median?

Answer

The median is 5. Put the numbers in order: 2, 4, 6, 8. With an even count, the median is the midpoint of the two middle numbers: $(4 + 6) \div 2 = 5$.

#406 · Number Sense and Operations

Write 75% as a decimal and as a fraction in lowest terms.

Answer

75% is 0.75 as a decimal and $\frac{3}{4}$ as a fraction. To change a percent to a decimal, divide by 100: $75 \div 100 = 0.75$. The decimal 0.75 is $\frac{75}{100}$, which lowers to $\frac{3}{4}$.

#407 · Geometry and Measurement

A triangle has a base of 12 cm and a height of 5 cm. What is its area?

Answer

The area is 30 square cm. Use one half x base x height: $0.5 \times 12 \times 5 = 30$. The base and height must be at a right angle to each other.

#408 · Data Management and Probability

The numbers are 3, 7, 7, 8, and 9. What is the mode?

Answer

The mode is 7. The mode is the value that appears most often. The number 7 appears twice and every other number appears once, so the mode is 7.

#409 · Number Sense and Operations

What is 20% of 150?

Answer

20% of 150 is 30. Change 20% to 0.20, then multiply: $0.20 \times 150 = 30$. You can also take one fifth of 150 because 20% is one fifth, and 150 divided by 5 = 30.

#410 · Geometry and Measurement

A circle has a radius of 7 cm. What is its circumference? Use $\pi = 22/7$.

Answer

The circumference is 44 cm. The formula is $2 \times \pi \times \text{radius}$, so $2 \times (22/7) \times 7$. The 7 in the denominator cancels the radius 7, leaving $2 \times 22 = 44$ cm.

#411 · Data Management and Probability

The numbers are 4, 9, 1, and 7. What is the range?

Answer

The range is 8. The range is the biggest value minus the smallest value. The largest is 9 and the smallest is 1, so $9 - 1 = 8$.

#412 · Number Sense and Operations

Work out $6 + 4 \times 2$ minus 3 using the order of operations.

Answer

The answer is 11. Do the multiplication first: $4 \times 2 = 8$. Then work left to right: $6 + 8 = 14$, then $14 - 3 = 11$. Multiplication comes before addition and subtraction.

#413 · Geometry and Measurement

A circle has a radius of 3 m. What is its area? Use $\pi = 3.14$.

Answer

The area is 28.26 square m. The formula is $\pi \times \text{radius squared}$, so $3.14 \times 3 \times 3$. Radius squared is 9 (3×3), and $3.14 \times 9 = 28.26$.

#414 · Data Management and Probability

The data set is 2, 4, 6, 8, and 10. How does adding the outlier 50 affect the mean?

Answer

Adding 50 pulls the mean up a lot. The original mean is 6 (30 divided by 5). With 50 included, the sum is 80 and the count is 6, so the mean becomes about 13.3. An outlier pulls the mean toward itself.

#415 · Number Sense and Operations

A jacket costs \$80 and is on sale for 30% off. What is the sale price?

Answer

The sale price is \$56. Find the discount: $0.30 \times 80 = \$24$. Subtract it from the original price: $80 \text{ minus } 24 = \$56$.

#416 · Geometry and Measurement

A right triangle has legs of 3 and 4. What is the hypotenuse?

Answer

The hypotenuse is 5. Use the Pythagorean theorem: $a^2 + b^2 = c^2$, so $9 + 16 = 25$. The square root of 25 is 5.

#417 · Data Management and Probability

Most house prices are close together but one house sells for a huge amount. Which measure best shows a typical price?

Answer

The median is the better measure. One extremely high price is an outlier that pulls the mean upward, so the mean stops representing the group. The median is the middle price and is not dragged by that one value.

#418 · Number Sense and Operations

A \$50 item carries 13% sales tax. What is the total cost?

Answer

The total is \$56.50. Find the tax: $0.13 \times 50 = \$6.50$.
Add it to the price: $50 + 6.50 = \$56.50$.

#419 · Geometry and Measurement

**A box is 4 cm long, 3 cm wide, and 2 cm tall.
What is its volume?**

Answer

The volume is 24 cubic cm. Multiply length x width x height: $4 \times 3 \times 2 = 24$. Volume is the space inside a 3D shape, so the units are cubed.

#420 · Data Management and Probability

On a bar graph, the bar for apples is twice as tall as the bar for oranges. What does this mean?

Answer

It means there are twice as many apples as oranges. In a bar graph, the height of each bar shows the amount for that category, so a taller bar always means a bigger value.

#421 · Number Sense and Operations

A restaurant bill is \$40 and you leave a 15% tip. How much is the tip?

Answer

The tip is \$6. Change 15% to 0.15 and multiply: $0.15 \times 40 = \$6$. You can also take 10% (\$4) plus half of that (\$2) to get \$6.

#422 · Geometry and Measurement

A cube has edges that are 5 cm long. What is its surface area?

Answer

The surface area is 150 square cm. One face has area $5 \times 5 = 25$. A cube has 6 faces, so $6 \times 25 = 150$. Surface area is the total area of all the outer faces.

#423 · Data Management and Probability

What does a histogram show, and why do its bars touch?

Answer

A histogram shows how often values fall into continuous intervals or bins. The bars touch because the intervals cover a continuous range with no gaps between them. This is different from a bar graph, which compares separate categories.

#424 · Number Sense and Operations

Write $17/5$ as a mixed number.

Answer

$17/5$ is 3 and $2/5$. Divide 17 by 5; 5 goes into 17 three times (15) with a remainder of 2. The whole part is 3 and the remainder 2 becomes the numerator over 5.

#425 · Geometry and Measurement

Convert 2.5 m to centimetres (cm).

Answer

2.5 m is 250 cm. There are 100 cm in 1 m, so multiply by 100: $2.5 \times 100 = 250$ cm. When moving from a larger unit to a smaller one, you multiply.

#426 · Data Management and Probability

A circle graph has sections of 25%, 40%, and 35%. Is this valid?

Answer

Yes, it is valid. A circle graph shows parts of one whole, so the percentages must add to 100%. Here $25 + 40 + 35 = 100\%$, so it divides the circle correctly.

#427 · Number Sense and Operations

A car travels 240 km in 3 hours. What is its unit rate in km per hour?

Answer

The unit rate is 80 km per hour. Divide the distance by the time: $240 \div 3 = 80$. A unit rate tells you the amount for one unit, here one hour.

#428 · Geometry and Measurement

Convert 3 L to millilitres (mL).

Answer

3 L is 3000 mL. There are 1000 mL in 1 L, so multiply by 1000: $3 \times 1000 = 3000$ mL. Litres and millilitres both measure volume.

#429 · Data Management and Probability

On a scatterplot, the points go up from the left toward the right. What does this tell you?

Answer

It shows a positive relationship. As the value on the horizontal axis increases, the value on the vertical axis also tends to increase. The points are read for the trend, not for a slope.

#430 · Number Sense and Operations

Solve the proportion $\frac{3}{4} = \frac{x}{12}$ for x .

Answer

$x = 9$. Cross multiply: $3 \times 12 = 4 \times x$, so $36 = 4x$.
Divide both sides by 4 to get $x = 9$.

#431 · Geometry and Measurement

A photo 4 cm wide is enlarged to 12 cm wide. What is the scale factor?

Answer

The scale factor is 3. Divide the new length by the original: $12 \div 4 = 3$. Every measurement on the enlarged copy is 3 times the original. Similar shapes share the same scale factor.

#432 · Data Management and Probability

One fair die is rolled. What is the probability of rolling a 4?

Answer

The probability is $\frac{1}{6}$. Probability is favorable outcomes divided by total outcomes. There is one 4 on the die and 6 sides in total, so 1 out of 6.

#433 · Number Sense and Operations

Work out $(-7) + 4$ and (-7) minus 4.

Answer

$(-7) + 4 = -3$, and (-7) minus 4 = -11. When adding a positive to a negative, subtract the smaller from the larger and keep the sign of the larger, so 7 minus 4 = 3 with a negative sign. When subtracting a positive, add the absolute values and keep the negative sign: $7 + 4 = 11$, so -11.

#434 · Geometry and Measurement

A cylinder has a radius of 2 cm and a height of 5 cm. What is its volume? Use $\pi = 3.14$.

Answer

The volume is 62.8 cubic cm. Use area of the base x height: $\pi \times \text{radius squared} \times \text{height} = 3.14 \times 4 \times 5$. Radius squared is 4, and $3.14 \times 4 \times 5 = 62.8$.

#435 · Data Management and Probability

A coin is tossed and a die is rolled. What is the probability of heads and a 6?

Answer

The probability is $\frac{1}{12}$. These are independent events, so multiply the chances. Heads is $\frac{1}{2}$ and rolling a 6 is $\frac{1}{6}$, and $\frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$.

#436 · Number Sense and Operations

A salesperson earns 8% commission on \$2,500 in sales. What is the commission?

Answer

The commission is \$200. Change 8% to 0.08 and multiply by the sales: $0.08 \times 2500 = \$200$. Commission is a percentage of the total sales amount.

#437 · Geometry and Measurement

An 8 m by 5 m rectangle has a 3 m by 2 m rectangle cut out of a corner. What is the remaining area?

Answer

The remaining area is 34 square m. The full rectangle is $8 \times 5 = 40$, and the cut-out is $3 \times 2 = 6$. Subtract them: $40 \text{ minus } 6 = 34$. For a composite shape, break it into parts and combine the areas.

#438 · Data Management and Probability

On one roll of a fair die, what is the probability of rolling a 2 or a 5?

Answer

The probability is $\frac{1}{3}$. A 2 and a 5 cannot both happen on one roll, so they are mutually exclusive. Add the chances: $\frac{1}{6} + \frac{1}{6} = \frac{2}{6}$, which lowers to $\frac{1}{3}$.

#439 · Number Sense and Operations

Find the simple interest on \$1,000 at 5% per year for 2 years.

Answer

The simple interest is \$100. Use $\text{interest} = \text{principal} \times \text{rate} \times \text{time}$. The rate as a decimal is 0.05, so $1000 \times 0.05 \times 2 = 100$. Simple interest is paid only on the original amount.

#440 · Geometry and Measurement

A square has an area of 49 square cm. How long is each side?

Answer

Each side is 7 cm. For a square, $\text{area} = \text{side} \times \text{side}$, so find a number times itself that equals 49. The square root of 49 is 7, because $7 \times 7 = 49$.

#441 · Data Management and Probability

The probability of rain is 0.3. What is the probability of no rain?

Answer

The probability of no rain is 0.7. Rain and no rain are complements that cover all outcomes. Subtract the given chance from 1: $1 - 0.3 = 0.7$.

#442 · Number Sense and Operations

A worker earns \$18 per hour and works 25 hours. What is the total pay?

Answer

The total pay is \$450. Multiply the hourly rate by the hours worked: $18 \times 25 = 450$. Total pay before any deductions is the hourly rate times the hours.

#443 • Geometry and Measurement

An L-shaped figure is made of a 4 cm by 6 cm rectangle and a 4 cm by 2 cm rectangle attached to its top right. What is the perimeter?

Answer

The perimeter is 28 cm. Walk the outside edges: 4 (bottom) + 4 (right of the lower part) + 4 (bottom of the top part) + 2 (right edge) + 8 (full top) + 6 (left edge) = 28 cm. Add every outside side once.

#444 • Data Management and Probability

A frequency table shows 5 students scored in the 80s and 7 students scored in the 90s. How many students are in the table?

Answer

There are 12 students. Add the frequencies for each row: $5 + 7 = 12$. A frequency table lists how many items or people fall into each category, and the total is the sum of all the frequencies.

#445 · Number Sense and Operations

What is 3.6 divided by 0.9?

Answer

The answer is 4. Move the decimal point one place in both numbers to clear the decimal divisor, giving 36 divided by 9. Then $36 \div 9 = 4$.

#446 · Geometry and Measurement

Convert 2500 g to kilograms (kg).

Answer

2500 g is 2.5 kg. There are 1000 g in 1 kg, so divide by 1000: $2500 \div 1000 = 2.5$. When moving from a smaller unit to a larger one, you divide.

#447 · Data Management and Probability

In a line, Tom is 3rd from the front and 5th from the back. How many people are in the line?

Answer

There are 7 people. Add the two positions and subtract 1, because Tom is counted in both: $3 + 5 = 8$, then $8 \text{ minus } 1 = 7$. It is a counting puzzle, not a probability question.

#448 · Number Sense and Operations

A price goes from \$40 to \$50. What is the percent increase?

Answer

The percent increase is 25%. Find the increase: $50 \text{ minus } 40 = 10$. Divide by the original amount: $10 \text{ divided by } 40 = 0.25$, which equals 25%. Always divide by the original value.

#449 · Geometry and Measurement

A right triangle has a hypotenuse of 13 and one leg of 5. What is the other leg?

Answer

The other leg is 12. Use the Pythagorean theorem:
 $13^2 - 5^2 = 169 - 25 = 144$.
The square root of 144 is 12.

#450 · Data Management and Probability

A spinner has 4 equal sections and 2 of them are red. What is $P(\text{red})$ as a percent?

Answer

$P(\text{red})$ is 50%. The chance is favorable divided by total: $2 \text{ out of } 4 = 0.5$. Change the decimal to a percent by multiplying by 100, so $0.5 = 50\%$.

#451 · Number Sense and Operations

What is $\frac{3}{4}$ written as a decimal?

Answer

The decimal is 0.75. Divide the numerator (top number) by the denominator (bottom number): 3 divided by 4 = 0.75.

#452 · Geometry and Measurement

**A rectangle is 8 cm long and 5 cm wide.
What is its area, in square centimetres?**

Answer

The area is 40 square cm. Area of a rectangle = length x width = $8 \times 5 = 40$. Area measures the flat space inside the shape.

#453 · Number Sense and Operations

What is 25% of 80?

Answer

The answer is 20. Write 25% as 0.25, then multiply:
 $0.25 \times 80 = 20$.

#454 · Geometry and Measurement

**A rectangle is 8 cm long and 5 cm wide.
What is its perimeter?**

Answer

The perimeter is 26 cm. Perimeter = $2 \times (\text{length} + \text{width}) = 2 \times (8 + 5) = 2 \times 13 = 26$. It is the total distance around the rectangle.

#455 · Number Sense and Operations

Simplify $5 + 3 \times 2$ using the order of operations.

Answer

The answer is 11. Do the multiplication first: $3 \times 2 = 6$, then add the 5: $5 + 6 = 11$. Remember BEDMAS, multiply and divide before you add and subtract.

#456 · Geometry and Measurement

A triangle has a base of 10 cm and a height of 6 cm. What is its area?

Answer

The area is 30 square cm. Area of a triangle = $(1/2) \times \text{base} \times \text{height} = (1/2) \times 10 \times 6 = 30$.

#457 · Number Sense and Operations

Write 0.6 as a fraction in lowest terms.

Answer

The fraction is $\frac{3}{5}$. Write 0.6 as $\frac{6}{10}$, then divide the top and bottom by 2: 6 divided by 2 = 3 and 10 divided by 2 = 5.

#458 · Geometry and Measurement

A square has a side length of 7 m. What is its perimeter?

Answer

The perimeter is 28 m. A square has 4 equal sides, so perimeter = $4 \times \text{side} = 4 \times 7 = 28$.

#459 · Number Sense and Operations

Write $\frac{3}{5}$ as a percent.

Answer

It is 60%. Divide the numerator by the denominator to get a decimal ($3 \text{ divided by } 5 = 0.6$), then multiply by 100: $0.6 \times 100 = 60\%$.

#460 · Geometry and Measurement

**A right triangle has legs of 3 cm and 4 cm.
What is the length of the hypotenuse?**

Answer

The hypotenuse is 5 cm. Use the Pythagorean theorem: $a^2 + b^2 = c^2$. So $3^2 + 4^2 = 9 + 16 = 25$, and the square root of 25 = 5.

#461 · Number Sense and Operations

Round 456.782 to the nearest tenth.

Answer

It rounds to 456.8. The tenths digit is 7 (the first digit after the decimal). Look at the next digit, 8, which is 5 or more, so the 7 rounds up to 8.

#462 · Geometry and Measurement

A circle has a radius of 5 cm. What is its area? Use $\pi = 3.14$.

Answer

The area is 78.5 square cm. Area of a circle = $\pi \times \text{radius squared} = 3.14 \times 5 \text{ squared} = 3.14 \times 25 = 78.5$.

#463 · Number Sense and Operations

Add: $\frac{2}{3} + \frac{1}{6}$.

Answer

The sum is $\frac{5}{6}$. Make the denominators the same. The lowest common denominator is 6, so rewrite $\frac{2}{3}$ as $\frac{4}{6}$. Then add the numerators: $\frac{4}{6} + \frac{1}{6} = \frac{5}{6}$.

#464 · Geometry and Measurement

A circle has a diameter of 10 cm. What is its circumference? Use $\pi = 3.14$.

Answer

The circumference is 31.4 cm. Circumference of a circle = $\pi \times \text{diameter} = 3.14 \times 10 = 31.4$.

#465 · Number Sense and Operations

Multiply: $\frac{2}{3} \times \frac{3}{4}$.

Answer

The product is $\frac{1}{2}$. Multiply the numerators ($2 \times 3 = 6$) and the denominators ($3 \times 4 = 12$) to get $\frac{6}{12}$. Then divide the top and bottom by 6 to get $\frac{1}{2}$.

#466 · Geometry and Measurement

A rectangular box is 4 cm long, 3 cm wide, and 2 cm high. What is its volume?

Answer

The volume is 24 cubic cm. Volume = length x width x height = $4 \times 3 \times 2 = 24$. Volume measures the space inside a 3D shape.

#467 · Number Sense and Operations

A jacket costs \$80 and is 25% off. What is the sale price?

Answer

The sale price is \$60. The discount is 25% of \$80, so $0.25 \times 80 = \$20$. Subtract the discount from the original price: $80 \text{ minus } 20 = \$60$.

#468 · Geometry and Measurement

A triangle has a base of 12 cm and an area of 36 square cm. What is its height?

Answer

The height is 6 cm. Area = $(1/2) \times \text{base} \times \text{height}$, so $36 = (1/2) \times 12 \times \text{height} = 6 \times \text{height}$. Divide 36 by 6 to get height = 6.

#469 · Number Sense and Operations

Solve for x: $x + 7 = 15$.

Answer

$x = 8$. Subtract 7 from both sides of the equation to undo the addition: $x = 15$ minus $7 = 8$.

#470 · Geometry and Measurement

How many millilitres are in 2.5 litres?

Answer

There are 2500 millilitres. There are 1000 millilitres in 1 litre, so multiply: $2.5 \times 1000 = 2500$.

#471 · Number Sense and Operations

Solve for x: $3x - 4 = 11$.

Answer

$x = 5$. Add 4 to both sides to get $3x = 15$, then divide both sides by 3: $x = 15$ divided by 3 = 5.

#472 · Geometry and Measurement

What is the volume of a cube with a side length of 3 cm?

Answer

The volume is 27 cubic cm. Volume of a cube = side x side x side = $3 \times 3 \times 3 = 27$.

#473 · Number Sense and Operations

Four pens cost \$6. What is the unit price for one pen?

Answer

The unit price is \$1.50. Divide the total cost by the number of items: $6 \div 4 = 1.5$, which is \$1.50.

#474 · Geometry and Measurement

What is the surface area of a cube with a side length of 3 cm?

Answer

The surface area is 54 square cm. A cube has 6 identical square faces. Each face is $3 \times 3 = 9$ square cm, so $6 \times 9 = 54$.

#475 · Number Sense and Operations

Solve for x: x divided by 5 = 8.

Answer

$x = 40$. Multiply both sides by 5 to undo the division:

$$x = 8 \times 5 = 40.$$

#476 · Geometry and Measurement

A model is built at a scale of 1 to 50. The real object is 5 m long. How long is the model?

Answer

The model is 10 cm long. Convert the real length to centimetres ($5 \text{ m} = 500 \text{ cm}$). At a 1 to 50 scale, divide by 50: $500 \text{ divided by } 50 = 10$.

#477 · Number Sense and Operations

Convert 2.5 kilograms to grams.

Answer

It is 2500 grams. There are 1000 grams in 1 kilogram, so multiply: $2.5 \times 1000 = 2500$.

#478 · Geometry and Measurement

A triangle has sides 3, 4, and 5. A similar triangle has sides 6, 8, and 10. What is the scale factor?

Answer

The scale factor is 2. Each side of the larger triangle is 2 times the matching side of the smaller one (6 divided by 3 = 2, 8 divided by 4 = 2, 10 divided by 5 = 2).

#479 · Number Sense and Operations

What is the simple interest on \$500 at 6% per year for 2 years?

Answer

The interest is \$60. Simple interest = principal x rate x time = $500 \times 0.06 \times 2 = 60$. First $500 \times 0.06 = 30$, then $30 \times 2 = 60$.

#480 · Geometry and Measurement

A right triangle has a hypotenuse of 13 cm and one leg of 5 cm. What is the other leg?

Answer

The other leg is 12 cm. Use the Pythagorean theorem: $c^2 - a^2 = b^2$. So $13^2 - 5^2 = 169 - 25 = 144$, and the square root of 144 = 12.

#481 · Number Sense and Operations

On a map, 1 cm represents 5 km. Two towns are 8 cm apart on the map. What is the real distance?

Answer

The real distance is 40 km. For every 1 cm on the map there are 5 km in real life, so multiply: $8 \times 5 = 40$ km.

#482 · Geometry and Measurement

A trapezoid has bases of 6 cm and 10 cm and a height of 4 cm. What is its area?

Answer

The area is 32 square cm. Area of a trapezoid = $(1/2) \times (\text{base 1} + \text{base 2}) \times \text{height} = (1/2) \times (6 + 10) \times 4 = (1/2) \times 16 \times 4 = 32$.

#483 · Number Sense and Operations

A car travels 240 km in 3 hours. What is its average speed?

Answer

The average speed is 80 km per hour. Speed equals distance divided by time: $240 \div 3 = 80$.

#484 · Geometry and Measurement

A parallelogram has a base of 8 cm and a height of 5 cm. What is its area?

Answer

The area is 40 square cm. Area of a parallelogram = base x height = $8 \times 5 = 40$. The height is the perpendicular distance from base to top, not the slanted side.

#485 · Number Sense and Operations

Write 0.75 as a percent.

Answer

It is 75%. Multiply the decimal by 100 to get a percent: $0.75 \times 100 = 75\%$.

#486 · Geometry and Measurement

A cylinder has a radius of 3 cm and a height of 4 cm. What is its volume? Use $\pi = 3.14$.

Answer

The volume is 113.04 cubic cm. Volume of a cylinder = $\pi \times \text{radius squared} \times \text{height} = 3.14 \times 3 \text{ squared} \times 4 = 3.14 \times 9 \times 4 = 3.14 \times 36 = 113.04$.

#487 · Number Sense and Operations

Simplify $10 - 8$ divided by 4 using the order of operations.

Answer

The answer is 8. Do the division first: 8 divided by $4 = 2$. Then subtract: 10 minus $2 = 8$.

#488 · Geometry and Measurement

A shape is a rectangle 6 cm by 4 cm with a triangle on top that has a base of 6 cm and a height of 3 cm. What is the total area?

Answer

The total area is 33 square cm. The rectangle is $6 \times 4 = 24$. The triangle is $(1/2) \times 6 \times 3 = 9$. Add them: $24 + 9 = 33$.

#489 · Number Sense and Operations

What is $\frac{2}{3}$ of 60?

Answer

It is 40. Multiply 60 by $\frac{2}{3}$: (60×2) divided by 3 = 120 divided by 3 = 40.

#490 · Geometry and Measurement

A rectangular prism is 4 cm long, 3 cm wide, and 2 cm high. What is its surface area?

Answer

The surface area is 52 square cm. Surface area = $2 \times (\text{length} \times \text{width} + \text{length} \times \text{height} + \text{width} \times \text{height})$
 $= 2 \times (12 + 8 + 6) = 2 \times 26 = 52.$

#491 · Number Sense and Operations

Solve for a: $5a + 12 = 22$.

Answer

$a = 2$. Subtract 12 from both sides to get $5a = 10$, then divide both sides by 5: $a = 10$ divided by 5 = 2.

#492 · Geometry and Measurement

Convert 3.5 metres to centimetres.

Answer

It is 350 centimetres. There are 100 centimetres in 1 metre, so multiply: $3.5 \times 100 = 350$.

#493 · Number Sense and Operations

A shirt costs \$24 and is on sale at 15% off. What is the discount amount?

Answer

The discount is \$3.60. Multiply the original price by the percent as a decimal: $0.15 \times 24 = 3.6$, which is \$3.60.

#494 · Geometry and Measurement

A circle has a radius of 4 cm. What is its circumference? Use $\pi = 3.14$.

Answer

The circumference is 25.12 cm. Circumference of a circle = $2 \times \pi \times \text{radius} = 2 \times 3.14 \times 4 = 25.12$.

#495 · Number Sense and Operations

Write $\frac{3}{8}$ as a decimal.

Answer

It is 0.375. Divide the numerator by the denominator: $3 \div 8 = 0.375$.

#496 · Geometry and Measurement

**A right triangle has legs of 3 cm and 4 cm.
What is its perimeter?**

Answer

The perimeter is 12 cm. Find the hypotenuse with the Pythagorean theorem: $3^2 + 4^2 = 25$, and the square root of $25 = 5$. Then add all three sides: $3 + 4 + 5 = 12$.

#497 · Number Sense and Operations

**A recipe uses 3 cups of flour for 4 servings.
How many cups are needed for 12
servings?**

Answer

You need 9 cups. Twelve servings is 3 times 4 servings, so multiply the flour by the same amount:
 $3 \times 3 = 9$ cups.

#498 · Geometry and Measurement

Convert 2500 grams to kilograms.

Answer

It is 2.5 kilograms. There are 1000 grams in 1 kilogram, so divide by 1000: 2500 divided by 1000 = 2.5.

#499 · Number Sense and Operations

Add: $-6 + 9$.

Answer

The sum is 3. Start at -6 on a number line and move 9 places up, which brings you to 3. It is the same as $9 \text{ minus } 6 = 3$.

#500 · Geometry and Measurement

A square has a side length of 9 cm. What is its area?

Answer

The area is 81 square cm. Area of a square = side x side = $9 \times 9 = 81$.